Check for updates

International Journal for Numerical Methods in Engineering

WILEY

RESEARCH ARTICLE

Time-Domain Spectral BFS Plate Element With Lobatto Basis for Wave Propagation Analysis

Hela Ambati[1] | Sascha Eisenträger[2] | Santosh Kapuria[1]

[1]Department of Applied Mechanics, Indian Institute of Technology Delhi, New Delhi, India | [2]Faculty of Mechanical Engineering, Otto von Guericke University Magdeburg, Magdeburg, Germany

Correspondence: Santosh Kapuria (kapuria@am.iitd.ac.in)

Received: 20 May 2024 | **Revised:** 10 September 2024 | **Accepted:** 22 October 2024

Funding: S. Kapuria thankfully acknowledges the financial support provided by the Science & Engineering Research Board, Department of Science and Technology, Government of India, through the J. C. Bose National Fellowship (Grant No. JBR/2023/000025) and the Core Research Grant (CRG/2021/001237).

Keywords: C^1-continuity | Kirchhoff plate theory | Lamb wave propagation | Lobatto basis | spectral BFS plate element | structural health monitoring

ABSTRACT

A computationally efficient spectral Kirchhoff plate element is presented for time-domain analysis of wave propagation at high frequencies in thin isotropic plates. It employs a C^1-continuous spectral interpolation based on the modified bi-Hermite polynomials using the Gauss–Lobatto–Legendre (GLL) points as a basis with selective collocation of rotational and twisting degrees of freedom (DOFs) at element edge and corner nodes. The lowest order version of the proposed element reduces to the classical Bogner–Fox–Schmit (BFS) element for Kirchhoff plates. The GLL basis allows diagonalisation of the mass matrix using the nodal quadrature technique, which enhances the computational efficiency. The numerical properties of the proposed element are comprehensively evaluated, including the conditioning of the system matrices. Moreover, the effect of employing different numerical integration schemes and nodal sets is examined in both static and free vibration analyses. The effectiveness of the proposed element in wave propagation problems is evaluated by comparing its performance to the converged solutions achieved using the BFS element with a very fine mesh. Results demonstrate that the current element, without and even with mass matrix diagonalisation delivers exceptional accuracy while also exhibiting faster convergence and enhanced computational efficiency than the existing Kirchhoff plate elements.

1 | Introduction

Guided wave propagation [1, 2] has become increasingly prominent as an effective tool for nondestructive testing (NDT) [3] and structural health monitoring (SHM) across various domains, including thin-walled structures such as ship hulls, aircraft fuselages, pipelines, and pressure vessels, [4–9] as well as for applications in impact response, [10] acoustics, [11–13] seismology, [14] and biomechanics [15, 16]. Consequently, significant attention has been directed toward developing numerical techniques for enhanced computational efficiency of ultrasonic guided wave propagation in beam, plate, and shell-type structures [17–19]. To detect both embedded and surface defects of small size, ultrasonic-guided waves play a crucial role in offline NDT and emerging online SHM techniques. They can do so by traveling long distances at a very short interval of time. This rapid detection is facilitated by a high propagation velocity, while low attenuation enables the coverage of large areas with a minimal amount of sensors [20]. Additionally, guided waves are easily excited and received by low-cost and low-energy piezoelectric transducers, increasing the area of application of guided wave-based techniques as well as their cost-effectiveness.

© 2024 John Wiley & Sons Ltd.

However, their application in SHM requires meeting specific requirements, including the use of high-frequency waves to ensure the half-wavelength is less than the defect size, typically ranging from 50 kHz to 1 MHz. Besides, short computational times are essential since SHM relies on model-based strategies, which entails solving the inverse problems for damage characterization and localization from measured sensor signals. Moreover, numerical simulations of guided waves are often essential for designing SHM systems due to the high costs associated with experimental investigations.

The classical finite element method (FEM), while widely employed to solve complex problems, exhibits limitations in addressing high-frequency guided wave propagation phenomena. These limitations include the introduction of numerical period elongation and amplitude decay, leading to errors associated with numerical dispersion and dissipation. Additionally, FEM necessitates a considerable number of low-order (linear, quadratic) elements in the mesh, typically, at least 20 finite elements per shortest wavelength are required, alongside a fine time discretization to achieve convergence of solutions [18, 21–23]. Consequently, when dealing with high-frequency waves utilized in SHM applications, the computational burden often becomes prohibitively large [5, 24]. As a result, extensive efforts have been dedicated to developing improved methods for accurately and efficiently simulating wave propagation in structures. Detailed examinations of these endeavors can be found in the comprehensive reviews provided in the literature [19, 25, 26]. Various studies have been conducted to address these challenges and to enhance the computational efficiency and accuracy of simulating high-frequency wave propagation phenomena.

To this end, various nonconventional and specialized FEM approaches have been developed. The semi-analytical finite element (SAFE) method is one of them, where the modeling of the cross-section is done using FEM and sinusoidal functions are used to express the displacement field analytically in the propagation direction. The SAFE method has been explored extensively in the literature [27–31]. A similar semi-analytical finite element, called the wave finite element method (WFEM), was introduced by Mead, combining periodic structure theory with FEM [32–34]. However, the WFEM assumes homogeneous material properties and a uniform cross-section along the longitudinal direction, limiting its applicability in many practical SHM scenarios despite its significant reduction in computational time.

The time-domain spectral element method (SEM), originally introduced by Patera [35], stands out as one of the most prevalent nonconventional techniques for addressing guided wave propagation solution challenges within SHM frameworks. Distinguished by its utilization of higher-order Lagrangian interpolation functions, SEM differs from conventional FEM through its strategic placement of internal nodes. Instead of employing evenly distributed internal nodes, SEM positions these nodes at the roots of orthogonal polynomials, such as Chebyshev and Lobatto polynomials [36, 37]. This departure from the classical placement of nodes results in the opportunity to employ the nodal quadrature technique, leading to a mass matrix formulation that is naturally diagonal (lumped). This feature is particularly important for explicit time integration schemes, enhancing their computational efficiency by orders of magnitude. However, it must be noted that depending on the element formulation, the mass matrix might be severely under-integrated [38]. SEM formulations have been tailored for various dimensional settings, encompassing one-dimensional (1D) bar structures, [36] two-dimensional (2D) plane strain/stress domains, [36] and three-dimensional (3D) solid configurations [39–41]. These formulations consistently satisfy C^0-continuity of displacements. Moreover, SEM variants have been devised to accommodate piezoelectric materials, facilitating the analysis of actuation, propagation, and sensing of Lamb waves within aluminum structures induced with piezoelectric transducers [41]. An anisotropic ansatz space was employed to model Lamb wave propagation in thin plates [42]. To this end, a spectral interpolation based on Gauss–Lobatto–Legendre (GLL) orthogonal polynomials is utilized for the in-plane directions, while for the out-of-plane (thickness) direction, linear shape functions are used. This approach is based on the ideas proposed by Düster, Bröker and Rank [43] for modeling thin-walled structures via the *p*-version of FEM. This technique reduces the computational costs significantly while retaining its accuracy. Subsequent extensions have adapted 3D SEM frameworks to model laminated composite structures [44, 45].

In an alternative methodology introduced by Doyle, [46, 47], the challenge of wave propagation analysis is tackled in the frequency domain through wavelet, Laplace, or Fourier transforms. This approach, commonly referred to as the frequency domain spectral element method (FDSEM), entails the derivation of dynamic stiffness matrices that are frequency-dependent for individual elements. FDSEM has found application in resolving one-dimensional (1D) wave propagation phenomena occurring in straight and curved beams made of isotropic and laminated composites [48–52]. The Development of frequency domain spectral elements has been rooted in the first-order shear deformation theory (FSDT), Euler-Bernoulli beam theory, refined third-order theory (TOT), and the efficient layerwise (zigzag) theory [53]. These theories necessitate either C^1-continuity or C^0-continuity for deflection. The FDSEM is used in plane strain two-dimensional (2D) domains, which are homogeneous and layered, [54–56] as well as composite plates, [57, 58] employing classical lamination theory (CLT), particularly when one direction remains unbounded, to tackle wave propagation problems. Unlike conventional FEM, FDSEM entails a significantly reduced problem size, necessitating only as many elements as there are distinct impedance substructures. However, the precise numerical computation of inverse Laplace transforms is computationally intensive for the systems featuring multiple DOFs, susceptible to wrap-around errors, and frequently unstable [47]. Moreover, FDSEM is limited in its amenability to domains of arbitrary shape or when multiple directions within the domain are bounded.

In wave propagation analysis, an approach known as the enriched finite element method (EFEM) has emerged. By incorporating specialized functions, the enrichment of Lagrangian interpolation functions is done. Melenk and Babuška [59] introduced the partition of unity FEM (PUFEM), used in solving high-frequency Helmholtz problems with the domain enriched entirely. This method found application in high-frequency wave scattering and resolving short-wave diffraction phenomena within 2D domains [60, 61]. A novel variation of EFEM, incorporating wave packet functions defined within local element

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

domains, satisfying the partition of unity condition, and employed as enrichment functions, has been proposed [62, 63]. This method has demonstrated accuracy and computational efficiency in wave propagation analysis of 1D and 2D problems. All these enriched formulations adhere to C^0-continuity.

Hrabok and Hrudey [64] gave a comprehensive understanding of the evolution of thin plate elements. One of the earliest plate elements is the Adini–Clough–Melosh (ACM) [65] element, based on the Kirchhoff theory of thin plates. The Kirchhoff theory requires C^1-continuity of the deflection, ensuring which poses a challenge, making the development of conforming C^1 elements difficult. The ACM element is a four-node quadrilateral element with each node having three DOFs, namely, the transverse deflection w and two rotational components θ_x and θ_y. The element is, however, nonconforming in nature, failing to ensure continuity of slope normal to its edges [66, 67]. Soon after its inception, various conforming triangular elements were promptly introduced [68–71]. Quadrilateral conforming elements were devised through the combination of triangular elements [72, 73]. Notably, a conforming 16-DOF element was proposed, wherein the twist w_{xy} is also incorporated as DOF [74]. This element, referred to as the Bogner–Fox–Schmit (BFS) element, leverages the product of 1D cubic Hermite interpolation polynomials in two directions. Zienkiewicz et al. [75] underscored the superior accuracy of the BFS element among rectangular elements tailored for thin plate problems. Gopalacharyulu [76] incorporated complete second-order derivatives in the BFS element, claiming faster convergence for stresses. Watkins [77] used blending function interpolation to construct new higher-order (quartic) conforming rectangular elements. The need for a structured mesh topology inherent to the derivation of these C^1-continuous rectangular elements limits their applicability to plates with rectangular sub-domains only. For addressing the need to broaden their scope to encompass thin plates of arbitrary shapes, the fictitious domain approaches (FDA) can be applied. The BFS element has been recently revitalized by combining it with FDA [78, 79], which circumvents its shortcomings for non-rectangular shapes. In recent years, the advent of isogeometric analysis (IGA) has also provided an opportunity to derive Kirchhoff-plate and Love-shell elements satisfying C^1-continuity for general shapes [80–82].

Because of the challenging nature of achieving C^1-continuity in classical thin plate theory-based quadrilateral elements, researchers shifted their focus to thick plate theory elements, which mandate solely C^0-continuity. Among these, the Reissner-Mindlin theory has gained traction because of its simplicity. However, in practice, C^0 elements face issues with shear locking, particularly when the plate thickness approaches infinitesimal values. To overcome this issue, researchers have developed several methods, including reduced integration, [83] selective reduced quadrature (SRQ), [84] mixed interpolation of tensorial components (MITC) [85], the discrete Kirchhoff technique, [86] and the discrete shear gap (DSG) method [87]. Cen and Shang [88] have provided a comprehensive exposition on the advancements made in Mindlin plate elements.

Arguably, because of the relative ease of satisfying C^0-continuity, the Reissner-Mindlin theory, also called first order shear deformation theory (FSDT), has been widely employed to develop time-domain spectral plate elements for wave propagation problems. Zrahia and Bar-Yoseph [89] constructed spectral plate elements on the basis of this theory using GLL functions. Kudela et al. [36] utilized the spectral elements using the Lobatto basis in plates made of composites. Brito and Sprague [90] studied the impact of various quadrature schemes on the solution of the FSDT-based Legendre spectral elements. It is found that mixed quadrature schemes are more effective at circumventing shear locking, where SRQ is used to evaluate the shear-stiffness matrix and nodal quadrature (GLL) for the bending-stiffness matrix. Based on the same theory, a high-order shell element was proposed by Dang-Trung, Yang, and Liu [91] using Chebyshev polynomials as basis functions, which eliminate spurious zero energy modes as well as shear locking, when the degree of the basis functions is increased.

Analyzing thin-walled structures employing elements based on 1D beam and 2D plate/shell theories (such as the Euler-Bernoulli beam theory, Kirchhoff plate theory, and FSDT), provides significant computational efficiency advantages compared to using 2D or 3D continuum-based elements as well as layerwise theory-based elements. The time-domain SFEs based on such theories have been employed for modeling composite plates featuring piezoelectric transducers for SHM applications [92–95]. However, all of the SFE formulations mentioned above are on the basis of C^0-continuity interpolations, whereas several classical and advanced theories, such as the classical plate theory, refined third-order theory (TOT), and the zigzag theories, [53, 95] require C^1-continuity for deflection. Recently, a few SFE formulations based on C^1-continuous interpolation functions have been presented based on the Euler-Bernoulli and zigzag theory for wave propagation analysis of beams [96–98] and based on the Kirchhoff theory for free vibration analysis for plates [99].

In this article, our aim is to conduct a thorough investigation into the numerical characteristics of the time-domain spectral Kirchhoff plate element designed for analyzing wave propagation in thin isotropic plates. A C^1-continuous interpolation scheme is developed for an $n \times n$-noded spectral element, using the modified bi-Hermite polynomials with selective collocation of rotational and twist DOFs at the element edge and corner nodes. The lowest order version (4-node) of this element reduces to the well-known 16-dof BFS element. The roots of the $(n-2)$th order Lobatto polynomial (GLL points) are chosen to position internal nodes in the element. It is known that using the nodal quadrature technique results in a diagonal mass matrix, enhancing the computational efficiency of time integration when employing the explicit central difference scheme. Therefore, the effect of employing a lumped mass matrix on the numerical behavior of the proposed plate element is also examined. Two methods for treating the zero masses for rotational DOFs in the lumped mass matrix are discussed and assessed. The numerical properties of the proposed element are comprehensively evaluated, including the conditioning of the system matrices. Subsequently, a focused analysis is conducted to examine the impact of different interpolation and integration schemes on static and free vibration analyses of the spectral plate element. The effectiveness of the proposed element in flexural wave propagation problems is evaluated by comparing its performance to the converged solutions achieved using the BFS element with a very fine mesh. Results demonstrate

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

that the current element delivers exceptional accuracy while also exhibiting faster convergence and enhanced computational efficiency than the existing Kirchhoff plate elements.

2 | Hamilton'S Principle for Kirchhoff Plate Theory

A thin rectangular plate of homogeneous and isotropic material under transverse loading is considered. The mid-plane of the plate ($z = 0$) is considered as the reference plane. Let u_x, u_y and w be the in-plane and transverse displacements. According to the Kirchhoff plate theory, the variations of the in-plane displacement vector $\mathbf{u}$ and the transverse displacement w at time t along the thickness direction z are approximated as

$$\boldsymbol{u}\,(x,y,z,t) = -z\,\boldsymbol{w}_{\boldsymbol{0}_d}\,(x,y,t), \quad w\,(x,y,z,t) = w_0\,(x,y,t) \tag{1}$$

with

$$\boldsymbol{u} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad \boldsymbol{w}_{\boldsymbol{0}_d} = \begin{bmatrix} w_{0,x} \\ w_{0,y} \end{bmatrix} \tag{2}$$

where a subscript comma followed by, say x, denotes partial differentiation with respect to x. Substituting Equation (1) into the linear strain-displacement relationship, the bending strains ϵ for small deformations can be expressed as

$$\boldsymbol{\epsilon} = z\boldsymbol{\kappa}^{\boldsymbol{0}} \tag{3}$$

where

$$\boldsymbol{\kappa}^{\boldsymbol{0}} = -\begin{bmatrix} w_{0,xx} \\ w_{0,yy} \\ 2w_{0,xy} \end{bmatrix} \tag{4}$$

The weak formulation for the equations of motion of the plate is obtained using Hamilton's principle as

$$\begin{aligned} &\int_A \left[\bar{I}\delta\boldsymbol{w}_{\boldsymbol{0}_d}^{\mathrm{T}}\ddot{\boldsymbol{w}}_{\boldsymbol{0}_d} + \hat{I}\delta w_0\ddot{w}_0 + \delta\boldsymbol{\kappa}^{\boldsymbol{0}^{\mathrm{T}}}\boldsymbol{M} - p_z\delta w_0\right]\mathrm{d}A \\ &\quad - \int_{\Gamma_\sigma}\left[-M_n\delta w_{0,n} + (M_{ns,s} + V_n)\,\delta w_0\right]\mathrm{d}\Gamma = 0, \quad \forall\,\delta w_0 \end{aligned} \tag{5}$$

where δ is the variational operator, ρ represents the material mass density, $p_z(x,y,t)$ is the normal force per unit area acting on the mid-surface of the plate, A denotes the surface area of the mid-plane of the plate, and Γ_σ denotes the boundary curve of the mid-plane of the plate with n and s as its normal and tangential directions, where stress resultants are prescribed. $\boldsymbol{M} = \begin{bmatrix} M_x & M_y & M_{xy} \end{bmatrix}^{\mathrm{T}}$ are bending stress resultants and $\boldsymbol{V} = \begin{bmatrix} V_x & V_y \end{bmatrix}^{\mathrm{T}}$ are shear stress resultants defined in terms of normal stresses σ_i and shear stresses τ_{ij} as

$$\boldsymbol{M} = \int_{\frac{-h}{2}}^{\frac{h}{2}} z\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}\mathrm{d}z \quad \text{and} \quad \boldsymbol{V} = \int_{\frac{-h}{2}}^{\frac{h}{2}}\begin{bmatrix} \tau_{xz} \\ \tau_{yz} \end{bmatrix}\mathrm{d}z \tag{6}$$

M_n, M_{ns}, and V_n are the bending, twisting, and shear stress resultants acting on the plate boundary with normal n. The inertia constants are given by $(\hat{I}, \bar{I}) = \int_{\frac{-h}{2}}^{\frac{h}{2}} \rho(1, z^2)\,\mathrm{d}z$.

Considering that Equation (5) holds well for arbitrary δw_0, both the area integral and the line integral must vanish independently. The line integrand yields the variationally consistent essential and associated natural boundary conditions. In the subsequent development, we will use w to mean w_0, for simplicity.

3 | Formulation for C^1-Continuous Spectral Plate Element

3.1 | Nodal Distribution

When employing a higher-order conventional element with equidistant nodes, specifically, $(\xi_i, \eta_i) = \frac{2(i-1)}{n-1} - 1,\ \forall\, i \in [1, 2, \ldots, n]$, the interpolation is susceptible to Runge's phenomenon with the increase in interpolation points [38, 100]. However, in spectral elements, clustering of nodes near the boundaries is done to mitigate this issue associated with high-order interpolations. This clustering is commonly obtained by selecting the zeros of orthogonal polynomials as the coordinates of nodes of the element [100]. Examples of such polynomials include the Legendre polynomial, Lobatto polynomial, and second-order Chebyshev polynomial. This approach also guarantees exponential convergence with the increase in polynomial order p ($\propto (1/p)^p$), known as spectral convergence. The present C^1-continuous spectral element employs GLL points as the local node base. The nodes of the 2D spectral element are placed using Lobatto roots on the interval $[-1, 1]$ in both directions in the natural coordinate system (ξ, η). The coordinates of nodes of a $n \times n$ noded spectral element are thus taken as

$$\begin{cases} \xi_1 = -1, \quad \xi_i = r_{i-1}, \quad \xi_n = 1 \\ \eta_1 = -1, \quad \eta_j = r_{j-1}, \quad \eta_n = 1, \quad i, j = 2, \ldots, n-2 \end{cases} \tag{7}$$

where $r_1, r_2, \ldots r_{n-2}$ denote the zeros of the Lobatto polynomials, $L_{n-2}(\xi)$ and $L_{n-2}(\eta)$, of $(n-2)$th order. The Lobatto polynomial $L_n(\xi)$ of order n is given as [100]

$$L_n(\xi) = \frac{1}{2^{n+1}(n+1)!}\,\frac{\mathrm{d}^{n+2}}{\mathrm{d}\xi^{n+2}}\,(\xi^2 - 1)^{n+1}, \qquad \forall n \geq 0 \tag{8}$$

The nodal coordinates (ξ_i) defined in Equation (8) can alternatively be obtained as the roots of the complete Lobatto polynomial $\hat{L}_n(\xi)$ of order n, given by

$$\hat{L}_n(\xi) = (1 - \xi^2)\,L_{n-2}(\xi), \quad \forall n \geq 2 \tag{9}$$

Similarly, the interpolation nodes for η_j can be obtained from $\hat{L}_n(\eta)$. The family of the Lobatto polynomials up to order 7 is detailed in Table 1.

The orthogonality property of the Lobatto polynomials gives

$$\int_{-1}^{1} L_i(\xi)(1 - \xi^2)L_j(\xi)\mathrm{d}\xi = \frac{2(i+1)(i+2)}{2i+3}\delta_{ij} \tag{10}$$

where δ_{ij} is the Kronecker delta.

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Lobatto polynomials.

$L_0 = 1$	$L_4 = \frac{15}{8}(21\xi^4 - 14\xi^2 + 1)$
$L_1 = 3\xi$	$L_5 = \frac{1}{8}(693\xi^4 - 630\xi^2 + 105)\xi$
$L_2 = \frac{3}{2}(5\xi^2 - 1)$	$L_6 = \frac{1}{16}(3003\xi^6 - 3465\xi^4 + 945\xi^2 - 35)$
$L_3 = \frac{5}{2}(7\xi^2 - 3)\xi$	$L_7 = \frac{9}{16}(715\xi^6 - 1001\xi^4 + 385\xi^2 - 35)\xi$

The Legendre polynomials $Le_n(\xi)$ can also be used to express the complete Lobatto polynomial [100]

$$\hat{L}_n(\xi) = (n-1)(Le_{n-2}(\xi) - \xi Le_{n-1}(\xi)) \tag{11}$$

where the nth order Legendre polynomial $Le_n(\xi)$ can be recursively computed as described below [100]

$$Le_n(\xi) = \frac{1}{n}\left[(2n-1)\xi Le_{n-1}(\xi) - (n-1)Le_{n-2}(\xi)\right], \qquad \forall n > 2 \tag{12}$$

The recursion begins with $n = 3$ where $Le_1 = 1$ and $Le_2 = \xi$. Employing the Newton-Raphson method, the roots of $\hat{L}_n(\xi)$ are determined numerically, where the estimate ${}^{j+1}\xi_i$ of the ith root ξ_i in the $(j+1)$th iteration is updated, based on its estimate ${}^{j}\xi_i$ from the previous iteration according to

$${}^{j+1}\xi_i = {}^{j}\xi_i - \frac{\hat{L}_n({}^{j}\xi_i)}{\hat{L}'_n({}^{j}\xi_i)}, \qquad \forall j \geq 0 \tag{13}$$

Using Equation (11), the derivative of $\hat{L}_n(\xi)$ is given by

$$\hat{L}'_n(\xi) = (n-1)\left[Le'_{n-2}(\xi) - \xi Le'_{n-1}(\xi) - Le_{n-1}(\xi)\right] \tag{14}$$

where the derivative of $Le_n(\xi)$ can also be recursively computed as [100]

$$Le'_n(\xi) = \frac{n}{1-\xi^2}(Le_{n-1}(\xi) - \xi Le_n(\xi)) \tag{15}$$

Substituting Equation (15) in (14) and using the recursion relation in Equation (12), $\hat{L}'_n$ simplifies to

$$\hat{L}'_n(\xi) = -n(n-1)Le_{n-1}(\xi) \tag{16}$$

Substituting Equations (11) and (16) into (13), ${}^{j+1}\xi_i$ can be directly computed using Legendre polynomials as

$${}^{j+1}\xi_i = {}^{j}\xi_i - \frac{{}^{j}\xi_i Le_{n-1}({}^{j}\xi_i) - Le_{n-2}({}^{j}\xi_i)}{nLe_{n-1}({}^{j}\xi_i)} \tag{17}$$

The initial guess ${}^{1}\xi_i$ is taken as the Gauss–Lobatto–Chebyshev (GLC) points, that is,

$${}^{1}\xi_i = \cos\left(\frac{(i-1)\pi}{n-1}\right) \tag{18}$$

The iteration continues until the absolute difference $\left|{}^{j+1}\xi_i - {}^{j}\xi_i\right| < \epsilon$, where ϵ is the machine tolerance set to 2.2204×10^{-16}. The GLL spacing for a 7×7-noded element is listed in Table 2 and depicted in Figure 1.

TABLE 2 | Roots (ξ_i/η_j) and corresponding weights (W_i/W_j) for GLL quadrature.

Node no. (i/j)	ξ_i/η_j	W_i/W_j
1	-1.0000000000000000	0.04761904761904762
2	−0.830223896278567	0.2768260473615659
3	−0.4688487934707142	0.4317453812098626
4	0.0000000000000000	0.4876190476190476
5	0.4688487934707142	0.4317453812098626
6	0.830223896278567	0.2768260473615659
7	1.0000000000000000	0.04761904761904762

3.2 | Generalized C^1-Continuous Shape Functions for Spectral Plate Element

Consider a $n \times n$-noded spectral plate element shown in Figure 1 of dimension $l_x \times l_y$ and thickness h having nodal coordinates (ξ_i, η_j) as GLL nodes as described in Section 3.1. The selective collocation of DOFs onto the element nodes as shown in Figure 2 is performed to obtain a conforming element and additionally reduce computation time [99]. At the corner nodes, four DOFs are defined, namely the deflection w, the rotations $\theta_x(= w_{,x})$, $\theta_y(= w_{,y})$, and the twist $\theta_{xy}(= w_{,xy})$ as per the 4-node BFS element. Along the edges normal to the x-direction, at the mid-side nodes, w and θ_x are specified, while w and θ_y are assigned along the edges normal to the y-axis. Only deflection w is defined for the inner nodes of the element. The relation between global and local coordinates is written as

$$x = x_0 + \frac{l_x}{2}\xi, \quad y = y_0 + \frac{l_y}{2}\eta$$
$$\frac{\mathrm{d}(\,)}{\mathrm{d}x} = \frac{2}{l_x}\frac{\mathrm{d}(\,)}{\mathrm{d}\xi} \quad \text{and} \quad \frac{\mathrm{d}(\,)}{\mathrm{d}y} = \frac{2}{l_y}\frac{\mathrm{d}(\,)}{\mathrm{d}\eta} \tag{19}$$

where $x_0 = \frac{x_1+x_n}{2}$ and $y_0 = \frac{y_1+y_n}{2}$.

The element displacement vector $\mathbf{U}^e$ is given by

$$\mathbf{U}^e = \begin{bmatrix} \boldsymbol{u}^e_{11} & \boldsymbol{u}^e_{21} \ldots & \boldsymbol{u}^e_{ij} \ldots & \boldsymbol{u}^e_{nn} \end{bmatrix}^{\mathrm{T}} \tag{20}$$

where

$$\boldsymbol{u}^e_{ij} = \begin{cases} [w_{ij} \quad \theta_{x_{ij}} \quad \theta_{y_{ij}} \quad \theta_{xy_{ij}}]^{\mathrm{T}}, & i,j = 1,n \\ [w_{ij} \quad \theta_{x_{ij}}]^{\mathrm{T}}, & i = 1,n;\ j = 2,\ldots,n-1 \\ [w_{ij} \quad \theta_{y_{ij}}]^{\mathrm{T}}, & i = 2,\ldots,n-1;\ j = 1,n \\ [w_{ij}]^{\mathrm{T}}, & \text{otherwise} \end{cases} \tag{21}$$

Using Equation (19), rotations θ_x and θ_y can be expressed as

$$\theta_x = w_{,x} = \frac{2}{l_x}w_{,\xi}, \quad \theta_y = w_{,y} = \frac{2}{l_y}w_{,\eta} \tag{22}$$

Let $w(x,y)$ be the transverse displacement and (i,j) be the location of the node on the plate element $(i = 1,2,3,\ldots,n$ and

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

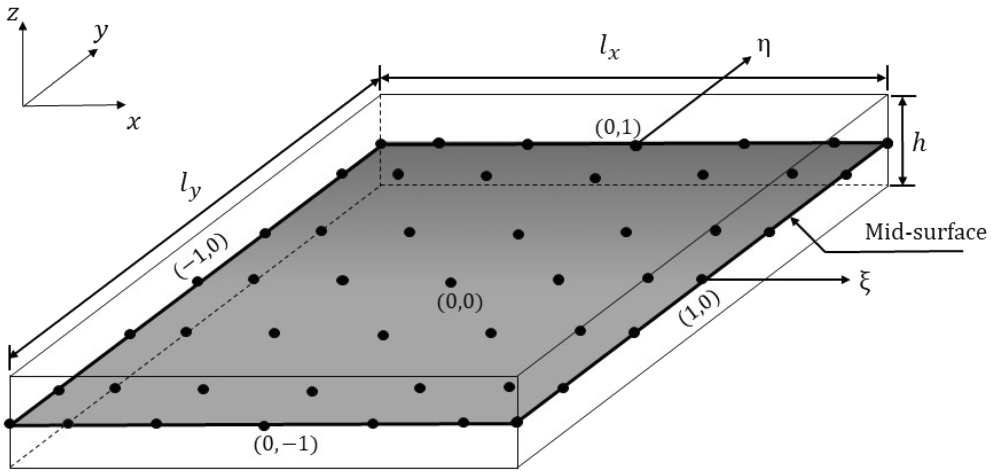


FIGURE 1 | Configuration of the rectangular plate element.

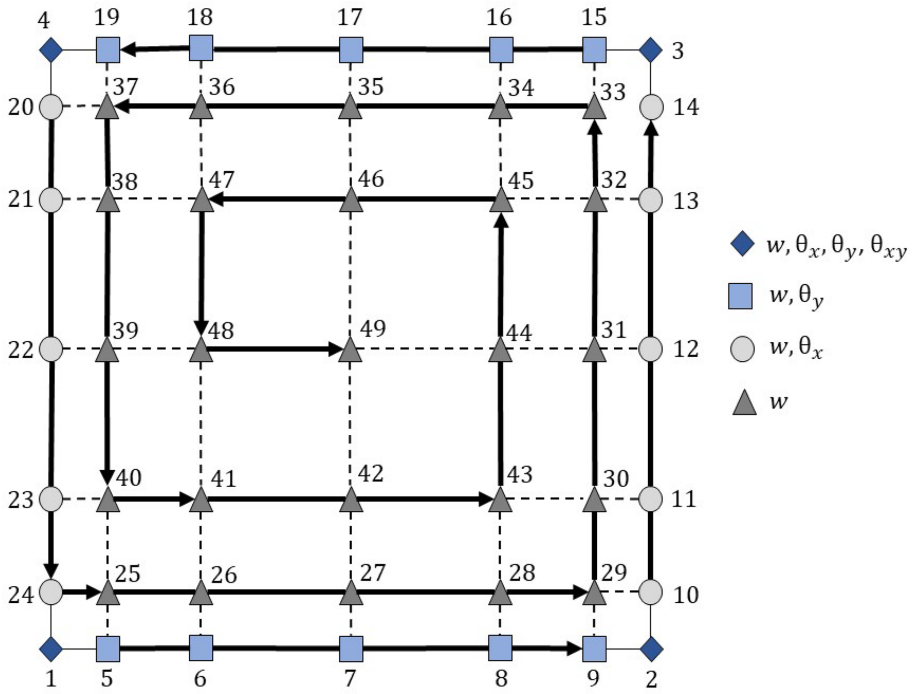


FIGURE 2 | Node distribution, numbering, and DOFs collocation of a 7 × 7-noded spectral element.

$j = 1, 2, 3, \ldots, n$). For a $n \times n$-noded element, w can be interpolated satisfying the continuity of first order derivatives of w (C^1-continuity) by employing the tensor product of 1D Hermite polynomials as

$$w(\xi,\eta) = \sum_{i=1}^{n}\sum_{j=1}^{n}\Big[H_{0i}^{(1)}(\xi)H_{0j}^{(1)}(\eta)w_{ij} + H_{1i}^{(1)}(\xi)H_{0j}^{(1)}(\eta)\theta_{x_{ij}} + H_{0i}^{(1)}(\xi)H_{1j}^{(1)}(\eta)\theta_{y_{ij}} + H_{1i}^{(1)}(\xi)H_{1j}^{(1)}(\eta)\theta_{xy_{ij}}\Big] \tag{23}$$

where $H_{ki}^{(N)}(s)$ denotes the Hermite polynomial that can interpolate up to the Nth derivative and is associated with the kth derivative of the variable w. Thus the superscript (1) indicates that the function w and its first derivative are matched at the interpolation points. These polynomials have the following properties at the interpolation points s_j ($j = 1, \ldots, n$):

$$\begin{aligned}
&\text{For } k = 0: H_{0i}^{(N)}(s_j) = \delta_{ij},\ \frac{\mathrm{d}H_{0i}^{(N)}(s_j)}{\mathrm{d}s} = 0, \ldots,\ \frac{\mathrm{d}^N H_{0i}^{(N)}(s_j)}{\mathrm{d}s^N} = 0\\
&\text{For } k = 1: H_{1i}^{(N)}(s_j) = 0,\ \frac{\mathrm{d}H_{1i}^{(N)}(s_j)}{\mathrm{d}s} = \delta_{ij}, \ldots,\ \frac{\mathrm{d}^N H_{1i}^{(N)}(s_j)}{\mathrm{d}s^N} = 0\\
&\vdots\\
&\text{For } k = N: H_{Ni}^{(N)}(s_j) = 0,\ \frac{\mathrm{d}H_{Ni}^{(N)}(s_j)}{\mathrm{d}s} = 0, \ldots,\ \frac{\mathrm{d}^N H_{Ni}^{(N)}(s_j)}{\mathrm{d}s^N} = \delta_{ij}
\end{aligned} \tag{24}$$

Note that the Hermite interpolation given by Equation (23) employs four DOFs w, θ_x, θ_y, and θ_{xy} for each of the $n \times n$ nodes in the element, yielding $4n^2$ DOFs per element. For the present spectral plate element, however, we eliminate the nonessential

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

derivatives of the noncorner and internal nodes as shown in Figure 2. We term the polynomials corresponding to this interpolation scheme as the modified Hermite polynomials. Considering $N = 1$ for the element C^1-continuous interpolation, we denote these polynomials by dropping the superscript (1) for simplicity as

$$\begin{aligned} H_{0i}^{(1)} &= H_{0i}, \quad \frac{\mathrm{d}}{\mathrm{d}\xi}\left(H_{0i}^{(1)}\right) = H'_{0i} \\ H_{1i}^{(1)} &= H_{1i}, \quad \text{and} \quad \frac{\mathrm{d}}{\mathrm{d}\xi_j}\left(H_{1i}^{(1)}\right) = H'_{1i} \end{aligned} \tag{25}$$

Equation (24) needs to be modified for this interpolation scheme to the following:

$$\begin{aligned} H_{0i}(\xi_j) &= \delta_{ij}, \quad i,j = 1,\ldots,n \\ H'_{0i}(\xi_j) &= 0, \quad i = 1,\ldots,n;\ j = 1,n \\ H_{1i}(\xi_j) &= 0, \quad i = 1,n;\ j = 1,\ldots,n \\ H'_{1i}(\xi_j) &= \delta_{ij}, \quad i,j = 1,n \end{aligned} \tag{26}$$

At locations ξ_j where function $H_{ki}(\xi)$ vanishes, it will have a root of multiplicity 1. On the other hand, if both $H_{ki}(\xi_j) = 0$ and $H'_{ki}(\xi_j) = 0$, it will have a root of multiplicity 2 at $\xi = \xi_j$. Thus, it follows from Equation (26) that ξ_1 is not a root of $H_{01}(\xi)$, $\xi_2, \xi_3, \ldots, \xi_{n-1}$ are its single roots, and ξ_n is a double root of $H_{01}(\xi)$. Therefore, $H_{01}(\xi)$ can be written in the form:

$$H_{01}(\xi) = (a\xi + b)(\xi - \xi_n)l_1(\xi) \tag{27}$$

where $l_i(\xi)$ denotes the Lagrange basis at $\xi = \xi_i$ given by

$$l_i(\xi) = \frac{(\xi-\xi_1)\cdots(\xi-\xi_{i-1})(\xi-\xi_{i+1})\cdots(\xi-\xi_n)}{(\xi_i-\xi_1)\cdots(\xi_i-\xi_{i-1})(\xi_i-\xi_{i+1})\cdots(\xi_i-\xi_n)} \tag{28}$$

The constants a and b are obtained by enforcing $H_{01}(\xi_1) = 1$ and $H'_{01}(\xi_1) = 0$ in Equation (27), using that $\xi_1 = -1$ and $\xi_n = 1$:

$$a = \frac{2l'_1(\xi_1) - 1}{4} \quad \text{and} \quad b = \frac{2l'_1(\xi_1) - 3}{4} \tag{29}$$

Substituting Equation (29) into Equation (27) gives

$$H_{01}(\xi) = (a_1\xi^2 + b_1\xi + c_1)l_1(\xi) \tag{30}$$

where

$$a_1 = \frac{2l'_1(\xi_1) - 1}{4}, \quad b_1 = -\frac{1}{2}, \quad \text{and} \quad c_1 = \frac{-2l'_1(\xi_1) + 3}{4} \tag{31}$$

Similarly, $H_{0n}(\xi)$ can be obtained as

$$H_{0n}(\xi) = (a_n\xi^2 + b_n\xi + c_n)l_n(\xi) \tag{32}$$

where

$$a_n = \frac{-2l'_n(\xi_1) - 1}{4}, \quad b_n = \frac{1}{2}, \quad \text{and} \quad c_n = \frac{2l'_n(\xi_n) + 3}{4} \tag{33}$$

For an intermediate node $i = 2,\ldots,n-1$, ξ_i is not a root, ξ_1 and ξ_n are double roots, and ξ_j $(j \neq i, 1, n)$ are single roots of $H_{0i}(\xi)$. Thus, $H_{0i}(\xi)$ for these intermediate nodes will take the form

$$H_{0i}(\xi) = a(\xi - \xi_1)(\xi - \xi_n)l_i(\xi) \equiv a(1 - \xi^2)l_i(\xi) \tag{34}$$

Enforcing that $H_{0i}(\xi_i) = 1$, the unknown constant a is obtained as

$$a = -\frac{1}{(1 - \xi_i^2)} \tag{35}$$

Thus, finally, the general expression for $H_{0i}(\xi)$ is obtained as

$$H_{0i}(\xi) = \frac{1 - \xi^2}{1 - \xi_i^2}l_i(\xi), \quad i = 2, \ldots.n-1 \tag{36}$$

From Equation (26), it is seen that $H_{11}(\xi)$ will have single roots at $\xi_1, \ldots, \xi_{n-1}$, a double root at ξ_n. Thus, $H_{11}(\xi)$ can be written as

$$H_{11}(\xi) = a(\xi - \xi_1)(\xi - \xi_2)\cdots(\xi - \xi_{n-1})(\xi - \xi_n)^2 \tag{37}$$

Applying the condition $H'_{11}(\xi_1) = 1$ into Equation (37), yields

$$a = \frac{1}{(\xi_1 - \xi_2)(\xi_1 - \xi_3)\cdots(\xi_1 - \xi_{n-1})(\xi_1 - \xi_n)^2} \tag{38}$$

resulting in

$$\begin{aligned} H_{11}(\xi) &= \frac{(\xi - \xi_1)(\xi - \xi_2)\cdots(\xi - \xi_{n-1})(\xi - \xi_n)^2}{(\xi_1 - \xi_2)(\xi_1 - \xi_3)\cdots(\xi_1 - \xi_{n-1})(\xi_1 - \xi_n)^2} \\ &= \frac{(\xi - \xi_1)(\xi - \xi_n)}{(\xi_1 - \xi_n)}l_1(\xi) \\ &= \frac{1 - \xi^2}{2}l_1(\xi) \end{aligned} \tag{39}$$

Similarly, we can determine H_{1n} as

$$H_{1n}(\xi) = \frac{\xi^2 - 1}{2}l_n(\xi) \tag{40}$$

Combining Equations (30), (32), (36), and (39), the modified Hermite polynomials of order $n+1$ for $\xi \in [-1, 1]$ are given by

$$\begin{aligned} H_{0i}(\xi) &= \begin{cases} (a_i\xi^2 + b_i\xi + c_i)\ l_i(\xi), & i = 1, n \\ \frac{1-\xi^2}{1-\xi_i^2}l_i(\xi), & i = 2, \ldots, n-1 \end{cases} \\ H_{1i}(\xi) &= -\frac{(1 - \xi^2)\xi_i}{2}l_i(\xi), \quad i = 1, n \end{aligned} \tag{41}$$

with

$$\begin{aligned} a_i &= \frac{-2\xi_i l'_i(\xi_i) - 1}{4}, \quad b_i = \frac{1}{2}\xi_i, \quad \text{and} \\ c_i &= \frac{2\xi_i l'_i(\xi_i) + 3}{4}, \qquad i = 1, n \end{aligned} \tag{42}$$

Applying Equation (23) to the current DOF collocation shown in Figure 2, the shape functions N_{ij}, N_{ij}^x, N_{ij}^y, and N_{ij}^{xy} associated

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

with the DOFs w_{ij}, $\theta_{x_{ij}}$, $\theta_{y_{ij}}$, and $\theta_{xy_{ij}}$, respectively, for node (i, j) at (ξ_i, η_j) are obtained as

$$\begin{aligned} N_{ij}(\xi,\eta) &= H_{0i}(\xi)H_{0j}(\eta), && i,j = 1,\ldots,n \\ N^x_{ij}(\xi,\eta) &= \frac{l_x}{2}H_{1i}(\xi)H_{0j}(\eta), && i = 1,n;\ j = 1,\ldots,n \\ N^y_{ij}(\xi,\eta) &= \frac{l_y}{2}H_{0i}(\xi)H_{1j}(\eta), && i = 1,\ldots,n;\ j = 1,n \\ N^{xy}_{ij}(\xi,\eta) &= \frac{l_x l_y}{4}H_{1i}(\xi)H_{1j}(\eta), && i,j = 1,n \end{aligned} \tag{43}$$

Note that for a four-node element with $n = 2$, Equation (41) leads to the standard cubic Hermite polynomials, and Equation (43) gives rise to the interpolation functions of the BFS element [74].

3.3 | Governing Equations of Motion

The displacement w and the curvature vector $\boldsymbol{\kappa_0}$ defined in Equation (4) can be expressed within a finite element as

$$w_0 = \mathbf{N}\mathbf{U}^e, \quad \delta w_0 = \mathbf{N}\,\delta\mathbf{U}^e, \quad \boldsymbol{\kappa_0} = \mathbf{B}\mathbf{U}^e, \quad \delta\boldsymbol{\kappa_0} = \mathbf{B}\,\delta\mathbf{U}^e \tag{44}$$

where $\mathbf{N} = \begin{bmatrix} \boldsymbol{N_{11}} & \boldsymbol{N_{21}} & \ldots & \boldsymbol{N_{ij}} & \ldots & \boldsymbol{N_{nn}} \end{bmatrix}$ and

$$\boldsymbol{N_{ij}} = \begin{cases} [N_{ij} \quad N^x_{ij} \quad N^y_{ij} \quad N^{xy}_{ij}], & i,j = 1,n \\ [N_{ij} \quad N^x_{ij}], & i = 1,n;\ j = 2,\ldots,n-1 \\ [N_{ij} \quad N^y_{ij}], & i = 2,\ldots,n-1;\ j = 1,n \\ [N_{ij}], & \text{otherwise} \end{cases} \tag{45}$$

Figure 2 illustrates the sequence of node numbering implemented for the 7×7-noded element in the solver. The elemental strain-displacement matrix $\mathbf{B}$ in Equation (44) is given as

$$\mathbf{B} = -\begin{bmatrix} \frac{\partial^2}{\partial x^2} \\ \frac{\partial^2}{\partial y^2} \\ 2\frac{\partial^2}{\partial x \partial y} \end{bmatrix}\mathbf{N} = -\begin{bmatrix} \frac{4}{l_x^2}\frac{\partial^2}{\partial \xi^2} \\ \frac{4}{l_y^2}\frac{\partial^2}{\partial \eta^2} \\ \frac{8}{l_x l_y}\frac{\partial^2}{\partial \xi \partial \eta} \end{bmatrix}\mathbf{N} \tag{46}$$

Substituting the reduced 3D constitutive equation for the state of plane stress in a plate, assuming negligible transverse normal stress σ_z and Equation (3) into (6), we obtain the plate constitutive equation as

$$\boldsymbol{M} = \mathbf{D}\boldsymbol{\kappa_0} \tag{47}$$

where

$$\mathbf{D} = \frac{Eh^3}{12(1-\nu^2)}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \tag{48}$$

The integral T in the variational Equation (5) for the entire domain is obtained by substituting the interpolations defined in Equation (44) along with Equation (47) into (5) for each element and summing up the contributions of n_e elements. Neglecting the rotary inertia terms considering thin plates, the integral T reduces to

$$\begin{aligned} T &= \sum_{e=1}^{n_e}\int_A \left[\delta\mathbf{U}^{e\mathrm{T}}\hat{I}\mathbf{N}^{\mathrm{T}}\mathbf{N}\ddot{\mathbf{U}}^e + \delta\mathbf{U}^{e\mathrm{T}}\mathbf{B}^{\mathrm{T}}\mathbf{D}\mathbf{B}\mathbf{U}^e - \delta\mathbf{U}^{e\mathrm{T}}\mathbf{N}^{\mathrm{T}}q\right]\mathrm{d}A \\ &\quad + \int_{\Gamma_\sigma}\delta\mathbf{U}^{e\mathrm{T}}\left(\left[\frac{\partial\mathbf{N}}{\partial n}\right]^{\mathrm{T}}_{\Gamma_\sigma}M_n + [\mathbf{N}]^{\mathrm{T}}_{\Gamma_\sigma}V_n\right)\mathrm{d}\Gamma \\ &= \sum_{e=1}^{n_e}\delta\mathbf{U}^{e\mathrm{T}}\left[\mathbf{M}^e\ddot{\mathbf{U}}^e + \mathbf{K}^e\mathbf{U}^e - \mathbf{F}^e_b - \mathbf{F}^e_\Gamma\right] \end{aligned} \tag{49}$$

where $\mathbf{K}^e$, $\mathbf{M}^e$, $\mathbf{F}^e_b$, and $\mathbf{F}^e_\Gamma$ are the elemental stiffness matrix, mass matrix, body force vector, and line load vector, respectively. They are given by

$$\begin{aligned} \mathbf{M}^e &= \int_{-1}^{1}\int_{-1}^{1}\rho h\mathbf{N}^{\mathrm{T}}\mathbf{N}\ \det\mathbf{J}\ \mathrm{d}\xi\mathrm{d}\eta \\ \mathbf{K}^e &= \int_{-1}^{1}\int_{-1}^{1}\mathbf{B}^{\mathrm{T}}\mathbf{D}\mathbf{B}\ \det\mathbf{J}\ \mathrm{d}\xi\mathrm{d}\eta \\ \mathbf{F}^e_b &= \int_{-1}^{1}\int_{-1}^{1}q\mathbf{N}^{\mathrm{T}}\ \det\mathbf{J}\ \mathrm{d}\xi\mathrm{d}\eta \\ \mathbf{F}^e_\Gamma &= \int_{\Gamma_\sigma}\left(\left[\frac{\partial\mathbf{N}}{\partial n}\right]^{\mathrm{T}}_{\Gamma_\sigma}M_n + [\mathbf{N}]^{\mathrm{T}}_{\Gamma_\sigma}V_n\right)\mathrm{d}\Gamma \end{aligned} \tag{50}$$

where n appearing in the expression of $\mathbf{F}^e_\Gamma$ takes the values of x and y, for which $\mathbf{N}$ is computed at Γ_σ defined by $\xi = \pm 1$ and $\eta = \pm 1$, respectively. The Jacobian matrix $\mathbf{J}$ is given by

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \frac{l_x}{2} & 0 \\ 0 & \frac{l_y}{2} \end{bmatrix} \tag{51}$$

It is important to note here that, for the rectangular element, the Jacobian matrix remains constant.

Summing of element contributions in Equation (49) is achieved by a process of assembly of the element matrices $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e (= \mathbf{F}^e_b + \mathbf{F}^e_\Gamma)$ into the corresponding global matrices $\mathbf{M}$, $\mathbf{K}$, and $\mathbf{F}$. This process yields the system equation of motion as

$$\mathbf{M}\ddot{\mathbf{U}} + \mathbf{K}\mathbf{U} = \mathbf{F} \tag{52}$$

where $\mathbf{U}$ represents the global vector of DOFs (deflections, rotations, and twists).

3.4 | GLL Quadrature

To compute the element matrices $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e$, we employ various quadrature methods. This section outlines the integration methods and the associated considerations. We use the GLL quadrature to calculate the integrals required for computing the element matrices $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e$ numerically. Using a k-point

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Lobatto quadrature formula, the method approximates the integral of a continuous function $f(\xi)$ given as

$$\int_{-1}^{1} f(\xi)\mathrm{d}\xi \approx \sum_{p=1}^{k} f(\xi_p)W_p, \quad k \geq 2 \tag{53}$$

where ξ_p are the roots of the k-degree complete Lobatto polynomial $\hat{L}_k$. W_p are the corresponding weights given by

$$\begin{aligned} W_1 = W_k &= \frac{2}{k(k-1)} \quad \text{and} \\ W_p &= \frac{2}{p(p-1)\,[Le_{p-1}(\xi_p)]^2}, \quad p = 2, 3, \ldots, k-1 \end{aligned} \tag{54}$$

The roots (ξ_p) and corresponding weights (W_p) of the 7-point GLL quadrature are shown in Table 2. The GLL quadrature gives an exact result of the integration of the function $f(\xi)$ only if it is a polynomial with a degree $2k-3$ or less. The degree of the interpolation functions for w_0 is $n+1$. Moreover, in the present study, simple geometries are used resulting in a constant Jacobian matrix. For our calculations involving the element matrices, the highest degree of polynomials in the integrands in the expressions of $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e$ are $2n+2$, $2n+2$ and $n+1$, respectively, assuming uniformly distributed load. Thus, for the exact integration of $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e$, the number of integration points required should meet the following condition:

$$\mathbf{K}^e : k \geq n + \frac{5}{2}, \qquad \mathbf{M}^e : k \geq n + \frac{5}{2}, \qquad \mathbf{F}^e : k \geq \frac{n+4}{2} \tag{55}$$

With the aid of quadrature formulae, the element matrices $\mathbf{K}^e$, $\mathbf{M}^e$, and $\mathbf{F}^e$ are numerically evaluated as shown in Equation (56), assuming that E, ρ, h, and q are constant. If these properties vary, we need to increase the number of integration points depending on the complexity of the distribution of those quantities to obtain accurate results.

$$\begin{aligned} \mathbf{K}^e &= \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^{\mathrm{T}}(\xi,\eta)\mathbf{D}\mathbf{B}(\xi,\eta)\det(\mathbf{J})\mathrm{d}\xi\mathrm{d}\eta \\ &= \sum_{i=1}^{n+3}\sum_{j=1}^{n+3} W_i W_j \mathbf{B}^{\mathrm{T}}\left(\xi_i,\eta_j\right)\mathbf{D}\mathbf{B}\left(\xi_i,\eta_j\right)\det(\mathbf{J}) \\ \mathbf{M}^e &= \int_{-1}^{1}\int_{-1}^{1} \rho h \mathbf{N}^{\mathrm{T}}(\xi,\eta)\mathbf{N}(\xi,\eta)\det(\mathbf{J})\mathrm{d}\xi\mathrm{d}\eta \\ &= \sum_{i=1}^{n+3}\sum_{j=1}^{n+3} W_i W_j \rho h \mathbf{N}^{\mathrm{T}}\left(\xi_i,\eta_j\right)\mathbf{N}\left(\xi_i,\eta_j\right)\det(\mathbf{J}) \\ &\quad \text{(consistent mass matrix)} \\ \mathbf{M}^e &\approx \sum_{i=1}^{n}\sum_{j=1}^{n} W_i W_j \rho h \mathbf{N}^{\mathrm{T}}\left(\xi_i,\eta_j\right)\mathbf{N}\left(\xi_i,\eta_j\right)\det(\mathbf{J}) \\ &\quad \text{(lumped mass matrix)} \\ \mathbf{F}_b^e &= \int_{-1}^{1}\int_{-1}^{1} q\mathbf{N}^{\mathrm{T}}(\xi,\eta)\det(\mathbf{J})\mathrm{d}\xi\mathrm{d}\eta \\ &= \sum_{i=1}^{\lceil \frac{n}{2} \rceil + 2}\sum_{j=1}^{\lceil \frac{n}{2} \rceil + 2} W_i W_j q \mathbf{N}^{\mathrm{T}}\left(\xi_i,\eta_j\right)\det(\mathbf{J}) \\ &\quad \lceil . \rceil = \text{Ceiling function} \end{aligned} \tag{56}$$

The process of mass lumping using the nodal quadrature technique ($n \times n$ integration points) involves selecting a common set of points to construct the polynomial interpolant and defining the quadrature rule, which serves as the basis. This approach ensures that off-diagonal terms in the elemental mass matrix become zero due to the orthogonality property of Lobatto polynomials shown in Equation (10), resulting in a lumped mass matrix for spectral elements. For the nodal quadrature method to be effective, the fundamental properties of interpolation functions like the partition of unity and Kronecker delta are crucial. If n-point integration is used for an n-node element, these characteristics ensure that only the diagonal components have nonzero values. Along with off-diagonal terms, the diagonal components related to rotational DOFs are zero, while nonzero values are only obtained for translational DOFs. Note that the GLL quadrature is used for all elemental matrices. However, to account for the different requirements (exact integration vs. mass lumping), the number of integration points is different for each quantity [101].

4 | Solution of the Equations of Motion

4.1 | Time Integration

An implicit direct time integration technique known as the Newmark-β method can be used for the semi-discrete equation of motion, as described in Equation (52). This method provides an unconditionally stable solution [102] and calculates the unknowns for the time step ($t + \Delta t$) from the previous time step (t):

$$\begin{aligned} &\left[a_1\,\mathbf{M} + a_2\,\mathbf{C} + \mathbf{K}\right]\mathbf{U}(t+\Delta t) \\ &\quad = \mathbf{P}(t+\Delta t) + \left[a_1\,\mathbf{M} + a_2\,\mathbf{C}\right]\mathbf{U}(t) \\ &\qquad + \left[a_3\,\mathbf{M} - a_4\,\mathbf{C}\right]\dot{\mathbf{U}}(t) + \left[a_5\,\mathbf{M} - a_6\,\mathbf{C}\right]\ddot{\mathbf{U}}(t) \end{aligned} \tag{57}$$

Here,

$$\begin{aligned} \ddot{\mathbf{U}}(t+\Delta t) &= a_1\,\{\mathbf{U}(t+\Delta t) - \mathbf{U}(t)\} - a_3\,\dot{\mathbf{U}}(t) - a_5\,\ddot{\mathbf{U}}(t) \\ \dot{\mathbf{U}}(t+\Delta t) &= a_2\,\{\mathbf{U}(t+\Delta t) - \mathbf{U}(t)\} + a_4\,\dot{\mathbf{U}}(t) + a_6\,\ddot{\mathbf{U}}(t) \end{aligned} \tag{58}$$

where $a_1 = 1/(\beta\Delta t^2)$, $a_2 = \gamma/(\beta\Delta t)$, $a_3 = 1/(\beta\Delta t)$, $a_4 = 1 - \gamma/\beta$, $a_5 = (1/2\beta) - 1$, $a_6 = 1 - (\gamma/2\beta)$, $\beta = \frac{1}{4}$, and $\gamma = \frac{1}{2}$. The central difference method, being a part of the Newmark family with $\gamma = 1/2$ and $\beta = 0$, is commonly employed when the mass matrix is diagonal $\mathbf{M}$ in the spectral formulation. In this explicit time integration scheme, the unknowns for the time step ($t + \Delta t$) are obtained as

$$\begin{aligned} &\left[a_1\,\mathbf{M} + a_2\,\mathbf{C}\right]\mathbf{U}(t+\Delta t) \\ &\quad = \mathbf{P}(t) + \left[2a_1\,\mathbf{M} - \mathbf{K}\right]\mathbf{U}(t) - \left[a_1\,\mathbf{M} - a_2\,\mathbf{C}\right]\mathbf{U}(t-\Delta t) \end{aligned} \tag{59}$$

where $a_1 = 1/\Delta t^2$ and $a_2 = 1/(2\Delta t)$.

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

In the systems with either a diagonal damping matrix $\mathbf{C}$ or $\mathbf{C}$ being zero (negligible damping), solving Equation (59) does not necessitate matrix inversion at each time step. To solve systems of equations with a large matrix size, this feature results in a substantial reduction in computational time.

4.2 | Methods for Handling Zero Masses

The lumped mass matrix $\mathbf{M}$ has zero diagonal terms corresponding to the rotational DOFs. In situations where there is no damping ($\mathbf{C} = 0$), it presents a challenge when solving Equation (59) directly using matrix inversion. However, this issue can be effectively addressed using two methods:

4.2.1 | Direct Explicit Method (DEM)

In the DEM, the mass matrix $\mathbf{M}$ is augmented by assigning a small value ϵ to its zero diagonal terms. This value ϵ is equal to the minimum value of nonzero diagonal terms (associated with translational DOFs), calculated as

$$\epsilon = \min(\mathbf{M}^e_{ii} \quad | \quad \mathbf{M}^e_{ii} \neq 0)/\gamma \tag{60}$$

Here, γ is a large constant, typically set to 10^7 [96]. The assembled mass matrix is then used to directly solve Equation (52) employing the explicit central difference method.

4.2.2 | Guyan Reduction Method (GRM)

Based on the zero and the nonzero terms in the diagonal mass matrix $\mathbf{M}$, the equations of motion are reorganized. Specifically, the equations are expressed in terms of two sets of DOFs:

1. Primary DOFs ($\mathbf{U}_p$): Corresponding to translational DOFs with nonzero diagonal terms in $\mathbf{M}$.
2. Secondary DOFs ($\mathbf{U}_s$): Representing rotational DOFs with zero diagonal terms in $\mathbf{M}$.

In this approach, we express Equation (52) solely in terms of the primary DOF vector $\mathbf{U}_p$ and employ the central difference method to solve this reduced system. Meanwhile, $\mathbf{U}_s$ is calculated by a derived relationship given in Equation (62). To this end, Equation (52) is partitioned and rearranged to distinguish between two sets of displacement DOFs:

$$\begin{bmatrix} \mathbf{M}_{pp} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{U}}_p \\ \ddot{\mathbf{U}}_s \end{bmatrix} + \begin{bmatrix} \mathbf{K}_{pp} & \mathbf{K}_{ps} \\ \mathbf{K}_{sp} & \mathbf{K}_{ss} \end{bmatrix} \begin{bmatrix} \mathbf{U}_p \\ \mathbf{U}_s \end{bmatrix} = \begin{bmatrix} \mathbf{P}_p \\ \mathbf{P}_s \end{bmatrix} \tag{61}$$

The equation for $\mathbf{U}_s$ in Equation (61) can be solved readily to obtain

$$\mathbf{U}_s = \mathbf{K}_{ss}^{-1}\mathbf{P}_s - \mathbf{K}_{ss}^{-1}\mathbf{K}_{sp}\mathbf{U}_p \tag{62}$$

Finally, we obtain the reduced system of equations of motion for $\mathbf{U}_p$ by eliminating $\mathbf{U}_s$ from Equation (61) using Equation (62) as

$$\mathbf{M}_{pp}\ddot{\mathbf{U}}_p + \overline{\mathbf{K}}_{pp}\mathbf{U}_p = \overline{\mathbf{P}}_p \tag{63}$$

where

$$\overline{\mathbf{K}}_{pp} = \mathbf{K}_{pp} - \mathbf{K}_{ps}\mathbf{K}_{ss}^{-1}\mathbf{K}_{sp}, \quad \overline{\mathbf{P}}_p = \mathbf{P}_p - \mathbf{K}_{ps}\mathbf{K}_{ss}^{-1}\mathbf{P}_s \tag{64}$$

Note that since the submatrices $\mathbf{M}_{ps}$, $\mathbf{M}_{sp}$, and $\mathbf{M}_{ss}$ of the mass matrix are all null matrices, GRM effectively reduces to static condensation and provides an exact solution of Equation (61).

Equation (63) is subsequently solved to determine the time history response of $\mathbf{U}_p$ by employing the central difference method. Concurrently, $\mathbf{U}_s$ is determined from Equation (62). Note that this approach is exact, as the condensed DOFs feature a zero mass term. Therefore, the Guyan reduction or static condensation can be used in this case without incurring the detrimental effects of introducing ϵ for the zero mass terms in the DEM. However, the GRM is computationally costly compared to the DEM because it requires additional initial matrix operations (see Equation (62)) and the inversion of a fully populated stiffness matrix (see Equation (64)).

5 | Numerical Results and Discussion

In this section, we first present the validation of the proposed element in comparison with analytical solutions available for static and free vibration responses of simply supported plates. The numerical properties of the proposed element, including the attainable rate of convergence and conditioning of the system matrices, are evaluated next. Further, the numerical study examines the impact of different interpolation and integration schemes of the spectral plate element. Finally, the computational advantage of the proposed element is illustrated for flexural wave propagation problems in comparison with other existing elements.

5.1 | Validation and Convergence Study

A MATLAB code has been developed for static, free vibration, and transient analysis of plates based on the proposed high-order spectral BFS element. The coding for the high-order BFS shape functions with selective collocation of DOFs is given in Appendix A. In order to validate the proposed formulation and the computer coding, we obtain the static bending and modal responses of an all-round, simply supported square plate, for which analytical solutions are available. These benchmark problems are first analyzed using the spectral plate element developed here employing GLL points in conjunction with a full integration (Equation (56)) of the system matrices $\mathbf{K}$ and $\mathbf{M}$ and the load vector $\mathbf{F}$. In the remainder of this paper, this approach is referred to as GLL-FI. This choice leads to a fully populated (consistent) elemental mass matrix; therefore, implicit time integration is used for transient and wave propagation problems in this case. The obtained results are compared to those from the spectral plate element presented by Wang et al. [99], who employed a quadrature rule based on GLC points, which uses only n points in each direction for an $n \times n$ node element. It is essentially the Clenshaw-Curtis quadrature rule, which is even less accurate compared to a n-point GLL quadrature. They applied this rule to $\mathbf{K}$, $\mathbf{M}$, and $\mathbf{F}$ and hence, under integrated all terms. Hence, they obtain a diagonal mass matrix by construction, but at the same time the stiffness matrix

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

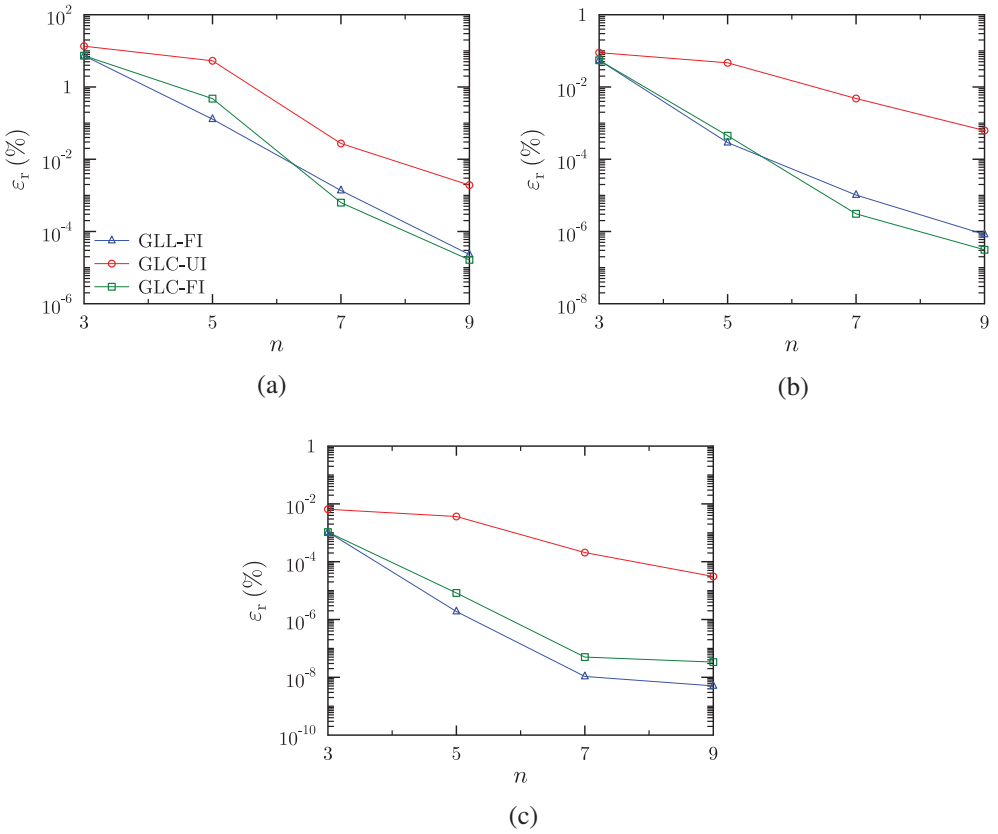


FIGURE 3 | Convergence of relative error in central deflection of different spectral plate element formulations (GLL-FI, GLC-UI [99], GLC-FI) with increasing number of nodes per edge of an element for meshes (a) 1×1, (b) 2×2, and (c) 4×4.

and the load vector are significantly under integrated. In the following, this approach is labelled GLC-UI.

To assess the relative performance of GLC basis functions in comparison with the GLL basis, another model is proposed, where GLC basis in combination with full integration is utilized. This scheme is referred to as GLC-FI. Using a unified reduced quadrature rule for both K and M matrices can lower the computational costs and ease the implementation into solvers, but at the cost of lower convergence rates. The stiffness matrix must be fully integrated, which will significantly improve the accuracy [101]. Additionally, the latter part of this section examines the rate of convergence considering static and modal analyses of the proposed element using both GLL and equidistant (EQ) points.

The geometric parameters of the plate are $L_x = L_y = 1\text{m}$, $h = 0.02$ m, and its material properties $E = 70\text{GPa}$ and $\nu = 0.3$, and $\rho = 2700$ kg/m^3. The plate is taken as an all-round simply supported so that the results obtained can be compared with analytical solutions. The plate is under a uniformly distributed load of $q_z = 1\text{kPa}$ in the static case. The error is computed for the deflection (w_c) at the center of the plate using the formula

$$\varepsilon_r = \left| \frac{w_c^{\text{num}} - w_c^*}{w_c^*} \right| \tag{65}$$

where w_c^{num} is the central deflection calculated numerically and w_c^* is the analytical solution [103]. For free vibration analysis, the first nine undamped natural frequencies are considered to obtain the root-mean-square error and are computed as

$$\epsilon_{\text{RMS}} = \sqrt{\frac{1}{9}\sum_{i=1}^{9}\left(\frac{\omega_i^{\text{num}} - \omega_i^*}{\omega_i^*}\right)^2} \tag{66}$$

where ω_i^{num} is the natural frequency of the ith mode determined through numerical techniques, and ω_i^* is its analytical solution from [104]. Results are obtained for the square plate with regular meshes of 1×1, 2×2, 3×3, and 4×4 and the number of nodes per edge of an element ranging from 3 to 9. Figures 3 and 4 illustrate the convergence characteristics of the three formulations, demonstrating much faster convergence of the current GLL-FI than the GLC-UI of Wang et al. [99]. The results obtained using GLC-FI too exhibit lesser accuracy compared to the GLL-FI, more so for the free vibration problem. The all-round simply-supported plate is again considered to plot frequency spectra using the proposed element (GLL-FI) which is compared with that of GLC-UI and GLC-FI for the 5×5-noded element, as shown in Figure 5a,b for two different regular meshes, 6×6 and 8×8, respectively. The natural frequencies ω_i^{num} computed numerically are normalized against analytical solutions ω_i^*, and plotted with respect to scaled mode number (i/N), where N represents the total number of modes ($N = 200$ in this case) and i ranges from 1 to 200. These numerical frequencies are arranged in ascending order.

The proposed element provides improved accuracy for both lower and higher modes compared to GLC-UI, particularly after mesh refinement, with deviations less than 1%. For these mesh sizes,

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

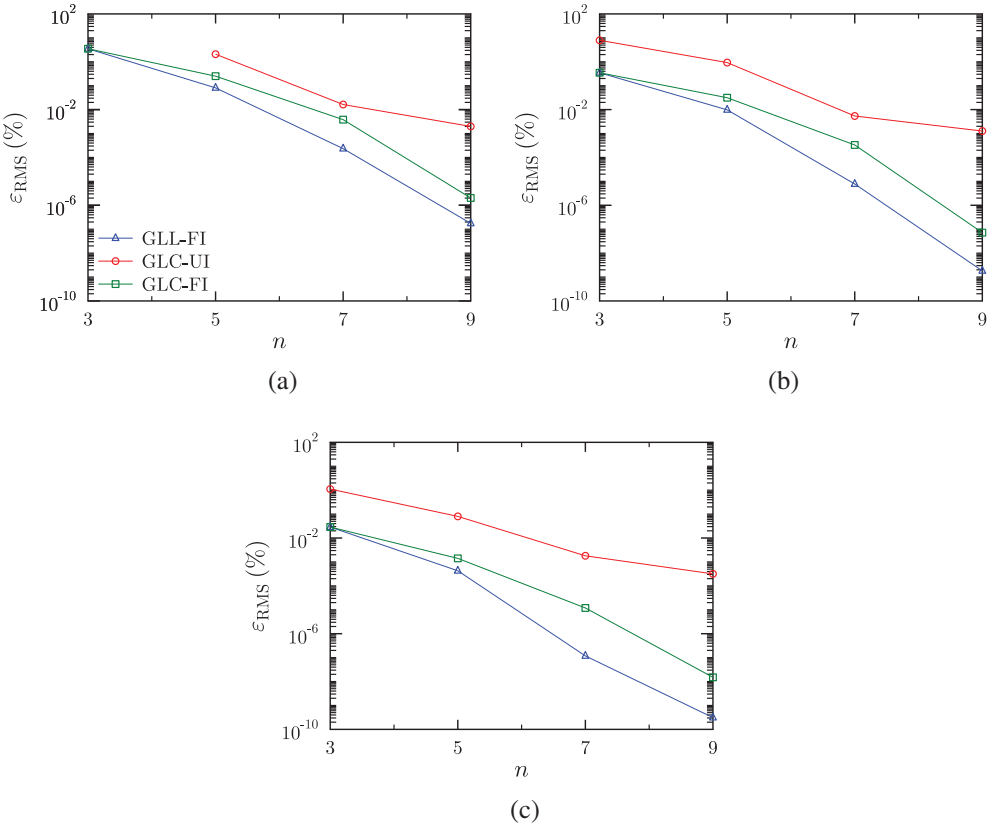


FIGURE 4 | Convergence of root-mean-square error in first nine natural frequencies of different spectral plate element formulations (GLL-FI, GLC-UI [99], GLC-FI) with increasing number of nodes per edge of an element for meshes (a) 2×2, (b) 3×3, and (c) 4×4.

there is not much difference seen in using GLL and GLC points in the case of full integration. Figure 5c illustrates the use of a lumped mass matrix to compute natural frequencies using the present element (referred to as GLL-UI*) with a regular mesh size of 8×8, while increasing the number of nodes per edge (p-extension). Note that GLL-UI* indicates that only the mass matrix is under integrated using the nodal quadrature technique to obtain the lumped mass matrix as given in Equation (56) while the other matrices are fully integrated. As the number of nodes per edge (n) of an element increases, the amplitudes of the oscillatory deviations from the analytical solution diminish, as depicted in Figure 5c,d. Note that there is, in fact, a dramatic increase in accuracy between 5 and 7 nodes over the edge, which is also observed later in Figure 7b. A similar trend is observed in Figure 5e with further mesh refinement (h-extension). Based on the analyses presented in Figures 3–5 with varying numbers of nodes per element edge, it is evident that the element with 7 nodes per edge achieves the optimal accuracy with the least number of DOFs. Consequently, we opt for a 7×7-noded spectral plate element for the subsequent wave propagation analysis.

The convergence plots for both static and free vibration problems of the simply supported plate problem described above are presented in Figures 6 and 7, respectively, for both GLL and EQ points. These figures illustrate the convergence behavior with respect to both p-extension (increasing the number of nodes per edge of an element from $n = 3$ to 9) and h-extension (doubling the elements for each simulation starting from one). It is to be noted that consistent mass formulation is utilized here, implying that any quadrature rule integrating the polynomials exactly will yield the same results. The standard Gaussian quadrature rule is applied for EQ points, while GLL quadrature is employed for GLL points. The plots reveal exponential convergence achieved with p-extension and algebraic convergence with h-refinement only for GLL-FI. In the case of static analysis, the nodal distribution did not significantly affect accuracy when a moderate number of nodes was used (up to $n = 5$). Conversely, in modal analysis, it is observed that GLL points exhibit superior convergence compared to EQ points in both p-extension and h-extension scenarios.

Next, we implement the present element using a lumped mass formulation (GLL-UI*) and examine the convergence behavior with the spatial discretization of 8×8 elements. In Figure 8, the exponential convergence is demonstrated as n is increased from 2 to 7. This includes the treatment of zero diagonal terms in the lumped mass matrix using both DEM and GRM as described in Section 4.2. In the case of DEM, the γ value considered is 10^{10}. Thus, the approximate DEM shows equally good performance as GRM.

5.2 | Condition Numbers of K and M Matrices

In this section, the conditioning of the matrices for the higher-order spectral plate element is examined. The condition number is a factor that has an inverse effect on the number of significant digits for direct solvers or the convergence rate of iterative

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

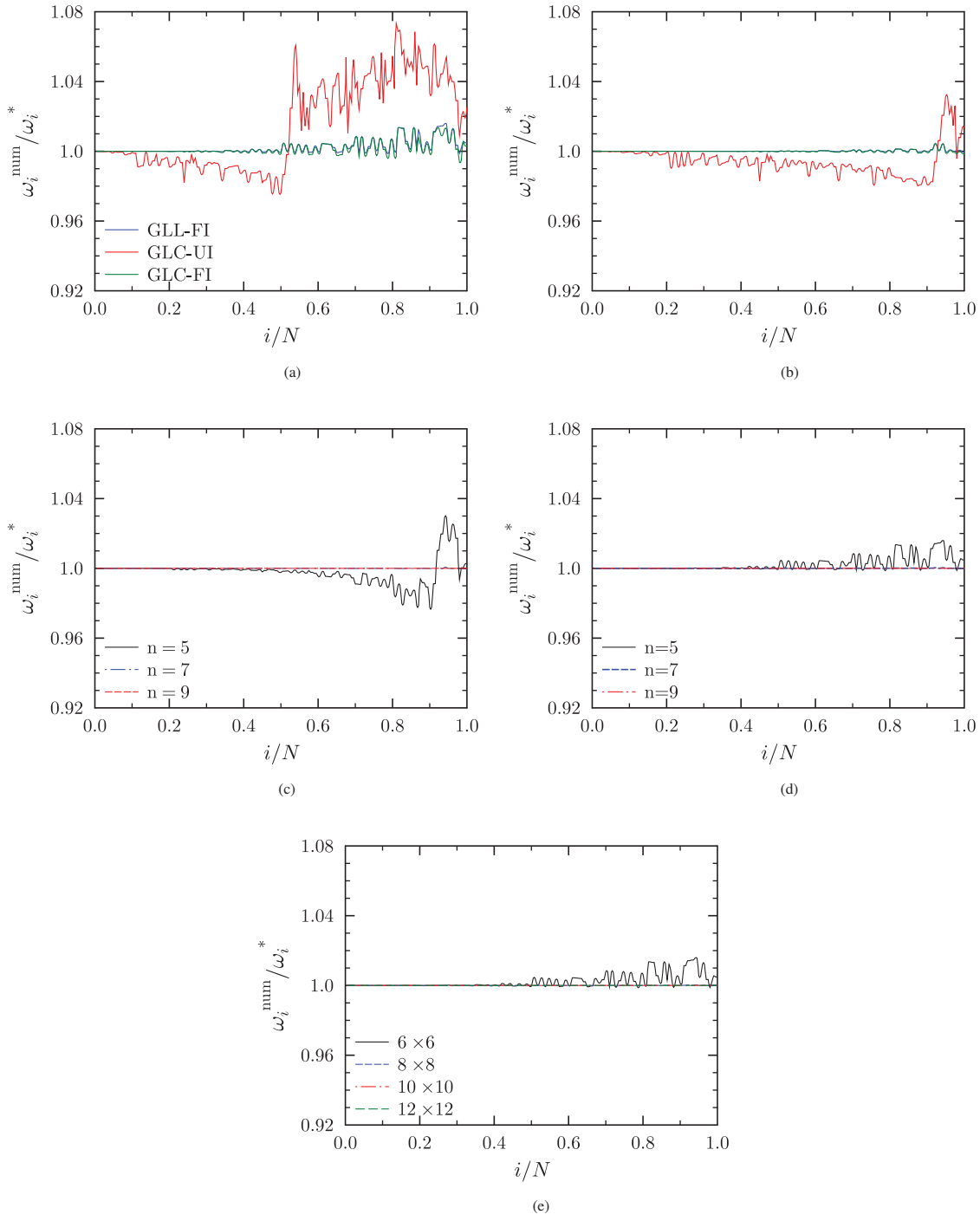


FIGURE 5 | Frequency spectra for the simply supported plate problem where frequencies normalized with respect to analytical solutions are plotted against scaled mode number for 200 flexural modes ($N = 200$). (a) Comparison of GLC-FI, GLC-UI [99], and GLC-FI for a mesh size of 6×6; (b) Comparison of GLC-FI, GLC-UI [99], and GLC-FI for a mesh size of 8×8; (c) Effect of n on error in frequencies using GLL-UI* with a mesh size of 8×8; (d) Effect of n on error in frequencies using GLL-FI with a mesh size of 6×6; (e) Effect of mesh size on error in frequencies using GLL-FI for $n = 5$.

solution methods. Understanding how the order of shape functions influences the condition numbers of matrices is pivotal for solving eigenvalue problems or linear systems of equations effectively. We investigate the variation of condition numbers κ for the stiffness matrix ($\mathbf{K}$), mass matrix ($\mathbf{M}$), and the product of the inverse mass matrix and stiffness matrix ($\mathbf{M}^{-1}\mathbf{K}$) concerning the polynomial order (p-extension) for both GLL points and EQ points. Examining the conditioning of $\mathbf{M}^{-1}\mathbf{K}$ is particularly crucial for ensuring stability in explicit transient analyses. The following expressions for the condition number $\kappa_{\mathbf{K}}$ of the stiffness matrices available in the literature [105–107] for C^0-continuous GLL and equidistant node-based Lagrange shape functions serve as a basis for the assessment of the conditioning of the present spectral plate element:

$$\kappa_{\mathbf{K}}^{\text{GLL}} = Ch^{-2}p^3 \tag{67}$$

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

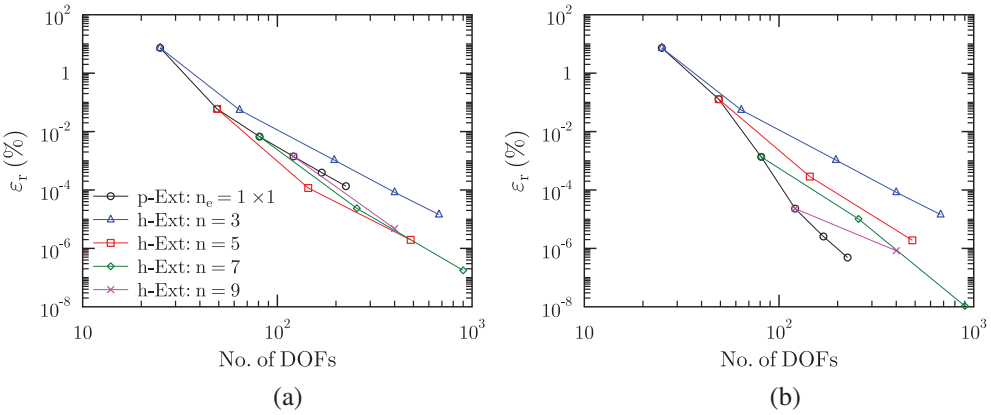


FIGURE 6 | Convergence curves in static analysis of simply supported plate under uniformly distributed load of 1 kPa for the present element (GLL-FI) with EQ and GLL nodes. (a) EQ nodes; (b) GLL nodes.

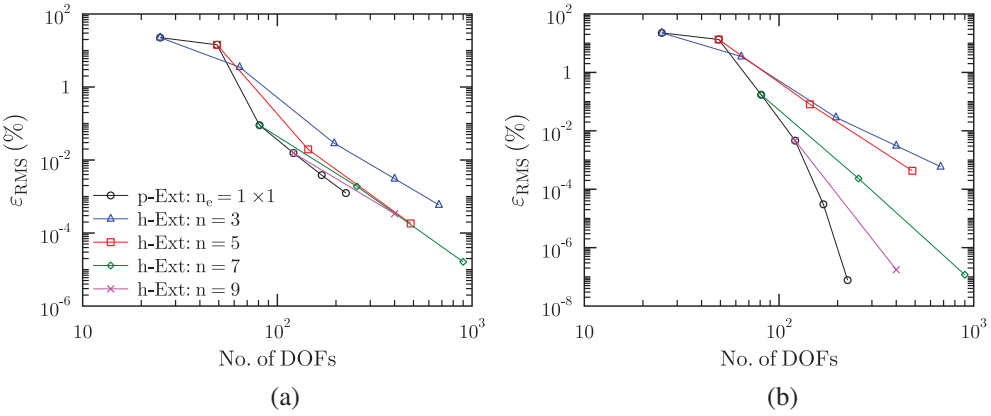


FIGURE 7 | Convergence for modal analysis of simply supported plate for the present element (GLL-FI) with EQ and GLL nodes. (a) EQ nodes; (b) GLL nodes.

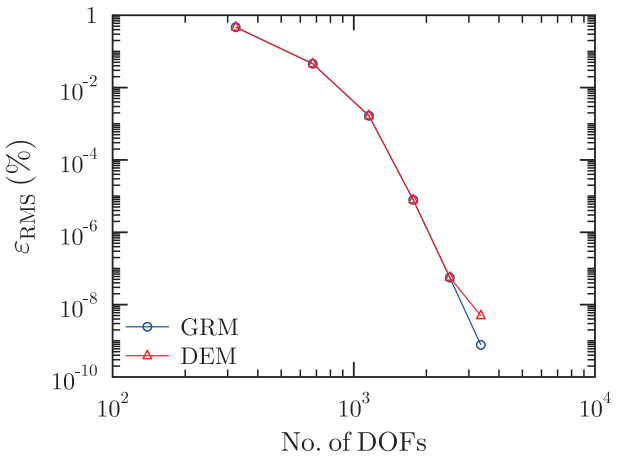


FIGURE 8 | Convergence curves for the spectral plate element in modal analysis of simply-supported plate using GRM and DEM techniques.

$$\kappa_{\mathbf{K}}^{\mathrm{EQ}} = Ch^{-2}p^{2-4d}4^{dp} \tag{68}$$

where h, p, and d denote the element size, polynomial order, and dimensionality of the problem. c is a constant. These equations can be generalized as shown in [38].

$$\kappa = c(h)p^{\alpha(d)} \tag{69}$$

where the constant c is a function of element size, α is a function of dimensionality d (in the present case $d = 2$). Using this generalized form, Equations (67) and (68) can be rewritten as

$$\log_{10}(\kappa_{\mathbf{K}}^{\mathrm{GLL}}) = c + \alpha \log_{10}(p) \tag{70}$$

with $c = Ch^{-2}$ and $\alpha = 3$ and

$$\log_{10}(\kappa_{\mathbf{K}}^{\mathrm{EQ}}) - dp\log_{10}(4) = c + \alpha \log_{10}(p) \tag{71}$$

with $c = Ch^{-2}$ and $\alpha = 2 - 4d$. To determine the suitability of the above relations based on the C^0-continuous shape functions to the current C^1-continuous shape functions, we fit the numerical data obtained in the plots of Figure 9 to Equation (70) for GLL points and Equation (71) for EQ points.

The condition number of the mass matrix using GLL C^0-continuous shape functions has been estimated in [107].

$$\kappa_{\mathbf{M}}^{\mathrm{GLL}} = Cp^{4d} \tag{72}$$

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

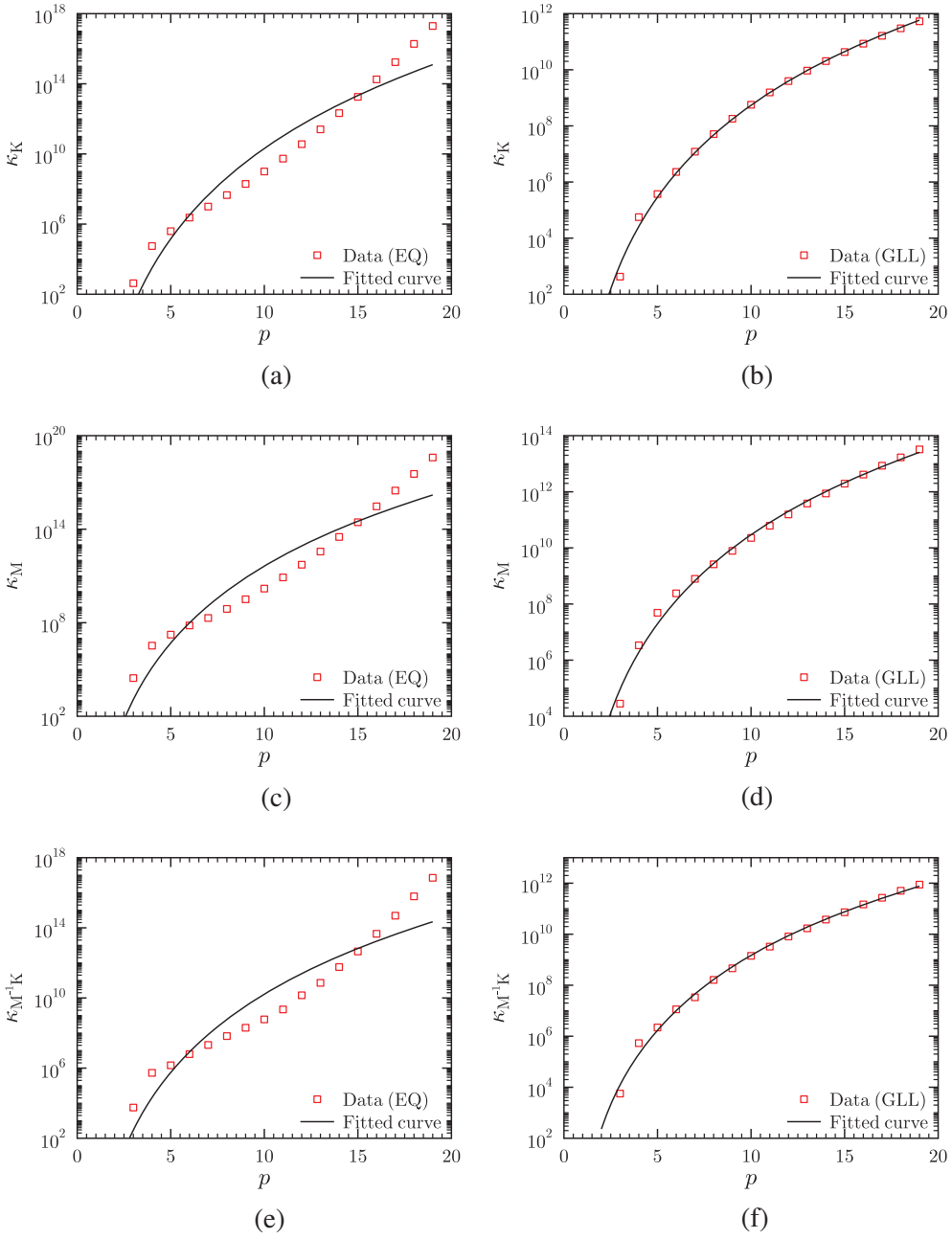


FIGURE 9 | Condition numbers of $\mathbf{K}$, $\mathbf{M}$, and $\mathbf{M}^{-1}\mathbf{K}$ matrices of present element against polynomial order of the interpolation functions using EQ or GLL points with a consistent mass matrix for a simply-supported plate. (a) $\kappa_{\mathbf{K}}^{EQ}$; (b) $\kappa_{\mathbf{K}}^{GLL}$; (c) $\kappa_{\mathbf{M}}^{EQ}$; (d) $\kappa_{\mathbf{M}}^{GLL}$; (e) $\kappa_{\mathbf{M}^{-1}\mathbf{K}}^{EQ}$; (f) $\kappa_{\mathbf{M}^{-1}\mathbf{K}}^{GLL}$.

Considering the similarity of the above estimates with Equation (69), the condition number of mass matrices is also fitted here using Equations (70) and (71).

An all-round simply supported square plate is discretized as a single spectral plate element, and the number of nodes per edge of the element (n) ranges from 2 to 18, with the order of the modified Hermite polynomials used to form shape functions $p = n + 1$. Note that in general the standard Hermite polynomials are of order $p = 2n - 1$. The data obtained for $\mathbf{M}$ in Figure 9 is using full numerical integration, which results in a consistent mass matrix. The choice of quadrature rule here can either be a standard Gaussian quadrature or a GLL quadrature with sufficient integration points, as mentioned in Section 3.4, since the data for conditioning remains the same for these matrices using both quadrature rules. Here, GLL quadrature is used for GLL points and standard Gauss quadrature for EQ points. It can be observed from Figure 9a,b that the trend of condition numbers of $\mathbf{K}$ for GLL points matches very well with the estimate given by Equation (70) with α-value of 10.874. The trend of $\kappa_{\mathbf{K}}$ for EQ points, however, does not follow the estimate given by Equation (71) so well. The α-value for this case is -7.537 as against the theoretical estimate of -6 for C^0-continuous Lagrange interpolations. A similar trend in the behavior of data and fitted curves as $\mathbf{K}$ is seen in $\mathbf{M}$ and $\mathbf{M}^{-1}\mathbf{K}$ for both GLL and EQ points. The α-value in Figure 9d for the $\mathbf{M}$ matrix is 10.542 for GLL points. Figure 9 also shows that

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

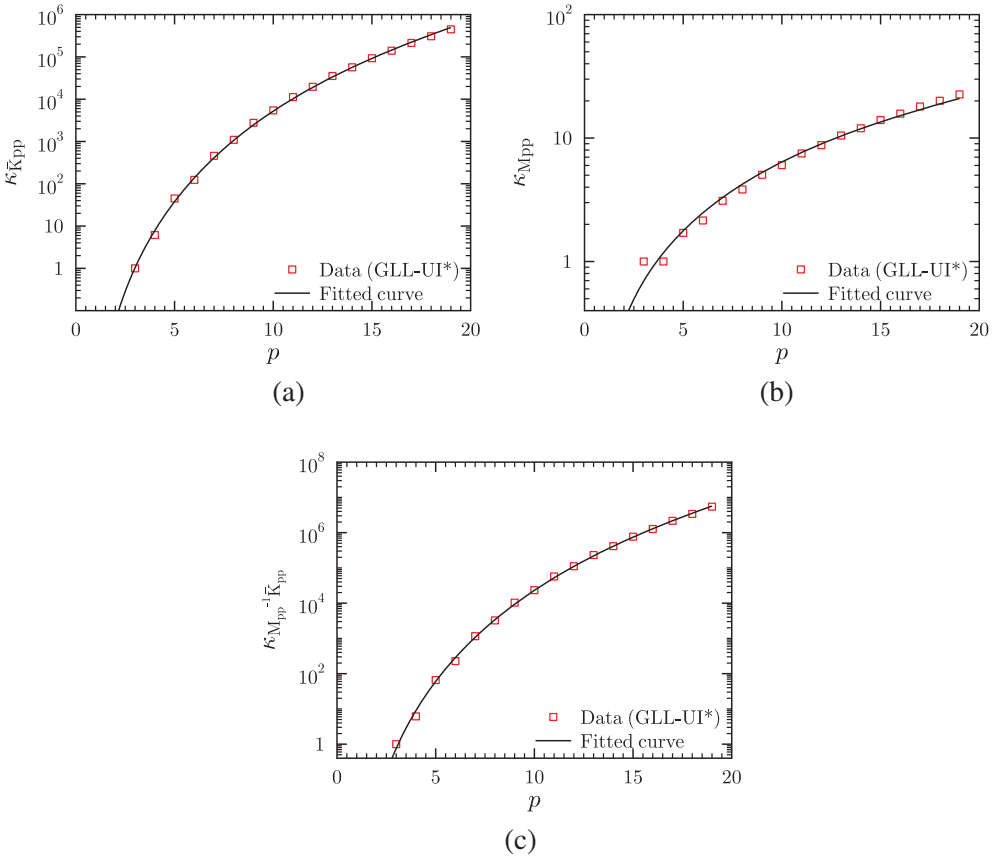


FIGURE 10 | Condition numbers $\overline{\mathbf{K}}_{\mathbf{pp}}$, $\mathbf{M}_{\mathbf{pp}}$, and $\mathbf{M}_{\mathbf{pp}}^{-1}\overline{\mathbf{K}}$ matrices of the present element against polynomial order of the interpolation functions using GLL points with a lumped mass matrix for a simply supported plate. (a) $\kappa_{\overline{\mathbf{K}}_{\mathbf{pp}}}$; (b) $\kappa_{\mathbf{M}_{\mathbf{pp}}}$; (c) $\kappa_{\mathbf{M}_{\mathbf{pp}}^{-1}\overline{\mathbf{K}}}$.

employing EQ points results in a severe increase in the condition number by several orders of magnitude for all the system matrices $\mathbf{K}$, $\mathbf{M}$, and $\mathbf{M}^{-1}\mathbf{K}$ with modified bi-Hermite interpolations. Thus, similar to C^0-continuous shape functions, for the C^1-continuous interpolations too, GLL points are advantageous over EQ points, not only by eliminating spurious oscillations at the ends, but also by yielding lower orders of condition number for all system matrices.

Furthermore, the conditioning of lumped mass matrices is also studied for the system matrices $\overline{\mathbf{K}}_{\mathbf{pp}}$, $\mathbf{M}_{\mathbf{pp}}$, and $\mathbf{M}_{\mathbf{pp}}^{-1}\overline{\mathbf{K}}$ in the reduced system of equations (63) obtained using GRM. For the simply supported plate, the discretization used here is 1×1 and the n ranges from 2 to 18, so that p varies from 3 to 19. Figure 10 illustrates that the orders of magnitude of the condition numbers are lower than those of consistent mass matrices depicted in Figure 9d,e. Consequently, these results indicate that the conditioning of system matrices using a lumped mass formulation is superior to that of the consistent mass formulation. The fitting data (α-values) for the system matrices for EQ and GLL points with full integration and for GLL-UI* are listed in Table 3.

TABLE 3 | Fitting data of $\mathbf{K}$, $\mathbf{M}$, and $\mathbf{M}^{-1}\mathbf{K}$ for EQ and GLL points in consistent mass and lumped mass formulation.

α	K	M	$M^{-1}K$
EQ	−7.537	−8.295	−9.851
GLL	10.87	10.54	9.736
GLL-UI* (GRM)	7.095	1.847	8.565

5.3 | Wave Propagation Analysis in Rectangular Plates

In this section, to further validate and assess the efficiency and accuracy of the proposed spectral plate element, wave propagation problems in a rectangular thin plate are investigated. These problems involve the actuation of piezoelectric transducers induced to the surface to excite Lamb waves in beam- and plate-like structures, which is of great interest in the field of SHM.

In the absence of any other published results in the literature for wave propagation analysis based on Kirchhoff's plate theory, for validation purposes, we consider the problem of straight-crested wave propagation in a panel of infinite width, previously addressed by Kapuria and Jain [96] using a time-domain C^1-continuous spectral Euler-Bernoulli beam element (SBE) under plane strain condition. The plate is assumed to be of infinite length, and the loading constant along the y-direction, leads to a plane strain condition in the $x - z$ plane. Note that the Euler–Bernoulli beam theory is the 1D version of the Kirchhoff plate theory formulation being employed here. The element is used to model a plate of 1.5 m length and 1.6 mm thickness as shown in Figure 11. The piezoelectric transducer

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

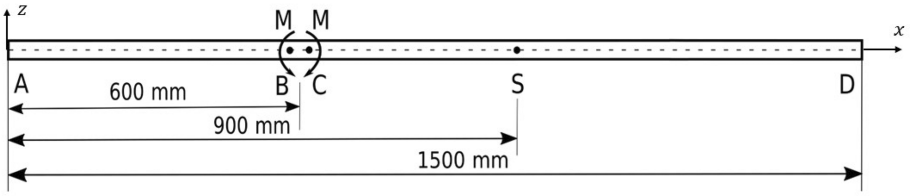


FIGURE 11 | Panel considered for validation of wave propagation analysis.

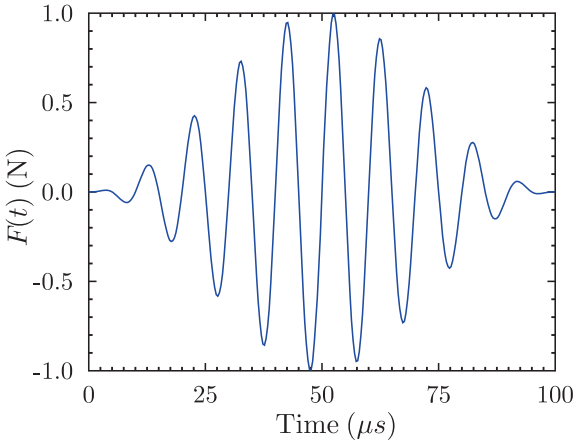


FIGURE 12 | Excitation signal with 100 kHz as central frequency.

has a width of 7 mm and is located between points B and C. On the top surface of the model at location BC, the actuator patch is bonded, while the response is recorded at the sensor location S. Using the pin-force method [108], the actuation of the piezoelectric transducer can be simulated by applying two equal and opposite shear forces $F(t)$ on the two edges of the patch actuator, parallel to the surface. The response is taken based on the flexural model using the SBE, where moments ($M(t) = F(t)h/2$) caused by the shear forces on the two edges of the patch at the reference axis of the panel are applied at the actuator edge locations B and C. At points B and C, the moments are applied using the Hann window-modulated sinusoidal tone burst signal, given as follows:

$$F(t) = \frac{F_0}{2}\left[1 - \cos\left(\frac{2\pi t}{T_H}\right)\right]\sin(2\pi f t) \tag{73}$$

where f represents the central frequency and T_H denotes the duration of the signal, which is obtained from the ratio of the number of the cycles (N_B) in the tone burst signal to the central frequency ($T_H = N_B/f$). In this problem, the structure is excited at 100 kHz with a 10-cycle signal of amplitude $F_0 = 1$ N/m as shown in Figure 12. The plate material has Young's modulus of 70 GPa, mass density ρ of 2700 kg/m^3, and Poisson's ratio ν of 0.3.

To simulate plane strain condition in the model based on the present plate element, a section of 1 m width is considered along the y-direction as shown in Figure 13, and a distributed moment with the amplitude function given by Equation (73) is applied uniformly across the entire width of the plate. The converged time-history response of the transverse displacement w at the sensor location S is obtained using 100 7-node SBEs. Note that in the SBE of Kapuria and Jain, [96] the standard Hermite polynomials with GLL points are used as interpolation functions. This response is compared in Figure 14 with the converged solution of the plate model obtained using the present formulation with 7×7-node elements of 100×101 mesh size. In the current study, the implicit time integration scheme is employed, which yields an unconditionally stable solution and provides fewer time steps compared to explicit methods. Thus the numerical computations are conducted employing the Newmark-β method with $\beta = \frac{1}{4}$ and $\gamma = \frac{1}{2}$, taking a time step of $\Delta T = 2 \times 10^{-7}$s, for a central frequency of 100 kHz for both the panel and the plate models. Remarkably, a close correspondence between the two results is observed, thereby validating the present formulation for transient analysis as well as the computer program developed.

Next, to assess the convergence behavior of the developed spectral plate element, an aluminum plate with a square-shaped actuator patch of size 7 mm × 7 mm bonded on the top surface is modeled. The problem represents real applications where finite-sized patch transducers are used. The plate dimensions are taken as the same as the previous model (Figure 15). Distributed moments $M_x(t)$ and $M_y(t)$ are applied at the actuator edges, where $M_x(t) = M_y(t) = F(t)h/2$. The excitation signal $F(t)$ adheres to the form outlined in Equation (73), with the total force on each actuator edge taken equal to 7 N. The load vectors for the distributed moments M_x and M_y applied at the concerned element edges are computed using Equation (50) with $M_n = \{M_x, M_y\}$ and $V_n = 0$. The response is monitored at a designated sensor location denoted as S, positioned at a distance of 300 mm from the actuator.

The time-history response of the transverse displacement w is computed for two distinct central frequencies, $f = 100$ kHz and 150 kHz, employing the Newmark-β method with corresponding time steps of $\Delta T = 2 \times 10^{-7}$s and $\Delta T = 1.5 \times 10^{-7}$s, respectively.

In order to assess the accuracy of the present solution, in the absence of an analytical solution, the reference solution is obtained using the 2×2-node BFS element, which is the lowest order version of the present spectral element. To achieve an accurate spatial resolution of propagating waves using the BFS element, it is imperative to employ a fine mesh comprising at least 20 elements across the shortest wavelength λ_{min} [22]. The minimum wavelength, λ_{min}, is determined by the transverse wave speed $c_T = \sqrt{\frac{\mu}{\rho}}$, defined as the lowest phase velocity, such that $\lambda_{min} = c_T/f$. For transient analysis, the time step Δt must be shorter than the time required for a wave with maximum velocity, here taken as $c_L = \sqrt{\frac{E(1-\nu)}{\rho(1+\nu)(1-2\nu)}}$, to traverse the distance l_e between adjacent nodes: $\Delta t = l_e/c_L$. Ensuring that l_e satisfies the mesh size criterion $l_e \leq \lambda/20$ also guarantees that the time step adheres to the condition of at least 20 steps per wave cycle, that is, $\Delta t \leq 1/20f$. The element size and time step chosen for all cases

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

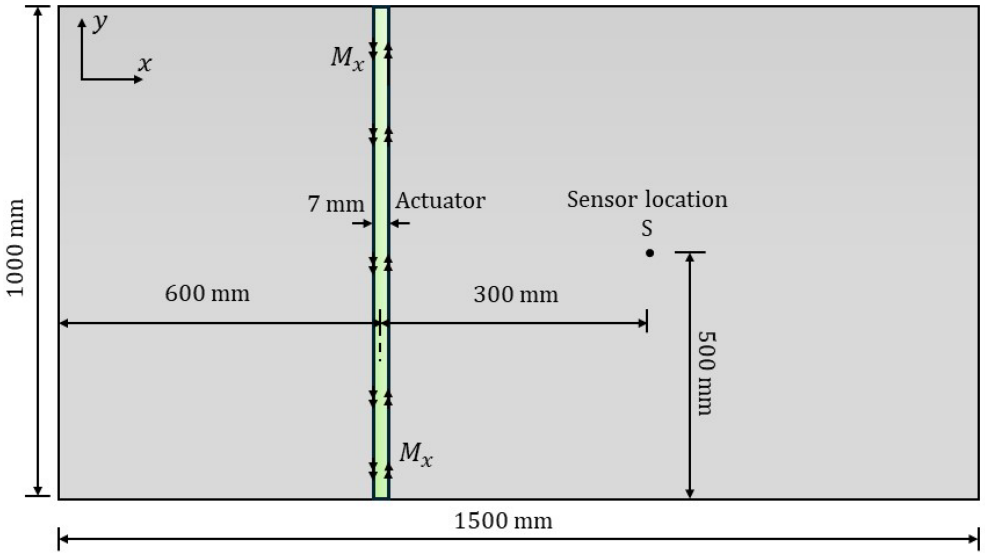


FIGURE 13 | Configuration of plate for validation.

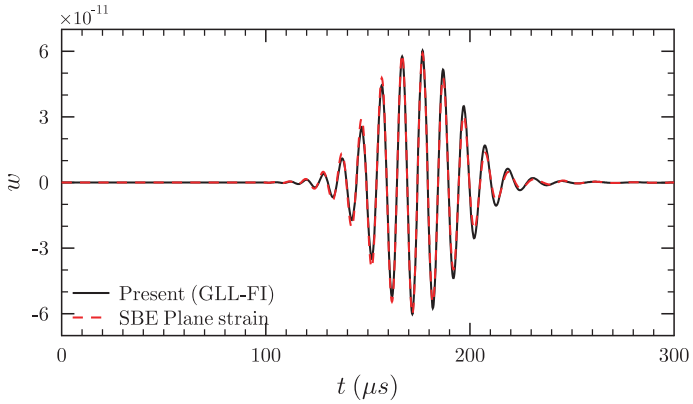


FIGURE 14 | Deflection-time-history response at location S of the plate under 10-cycle tone burst excitation along the width of the plate for excitation frequency 100 kHz with the reference problem using the TDSE element under plane strain condition [96].

in this study satisfy these criteria. Thus, the reference BFS element solution is established by considering a very fine mesh comprising 1,938,420 DOFs, employing the Newmark-β scheme with parameters as previously stated. Employing parallelization techniques for assembling the system matrices and saving the LU or Cholesky decomposition of the effective stiffness matrix to reuse in all steps during the time integration procedure proves to be highly effective in significantly reducing computation time [109, 110]. The transverse displacement responses recorded at location S for central frequencies of 100 and 150 kHz are illustrated in Figure 16, comparing the current formulation with the reference solution. Additionally, the response obtained using the GLC-UI element developed by Wang et al. [99] using the same mesh as in the present model for a meaningful comparison and the BFS element with the equivalent number of DOFs as the present model are also compared in Figure 16.

Further, the time-history response of transverse displacement of the same problem is computed at central frequencies, $f = 100$ and 150 kHz, with a 5-cycle signal using GLL-UI* shown in Figure 17 and is compared with that of GLL-FI given in Figure 16.

For all the examples presented, the implicit version of the Newmark method is used if the mass matrix is consistent, while the central difference method (CDM) is used for the lumped mass matrix. Here, CDM is employed using DEM for GLL-UI* with a time step of $\Delta T = 7 \times 10^{-9}$s where a stable solution is obtained. It may be preferable to use DEM over GRM due to its initial matrix operations, which consume a lot of computational time due to the large size of the matrices. The computational time is reduced by 5–6 times using DEM compared to GRM in the current example.

A comparative analysis of all the aforementioned solutions is presented in Table 4 in terms of the number of DOFs and percentage root-mean-square-deviation (RMSD). The RMSD is calculated using

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^{N_t}\left[w_{\text{num}}(t_i) - w_{\text{ref}}(t_i)\right]^2}{\sum_{i=1}^{N_t} w_{\text{ref}}^2(t_i)}} \tag{74}$$

where N_t denotes total number of time steps ($N_t = T/\Delta t$), $w_{\text{num}}(t_i)$ and $w_{\text{ref}}(t_i)$ are the deflection response at ith time

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

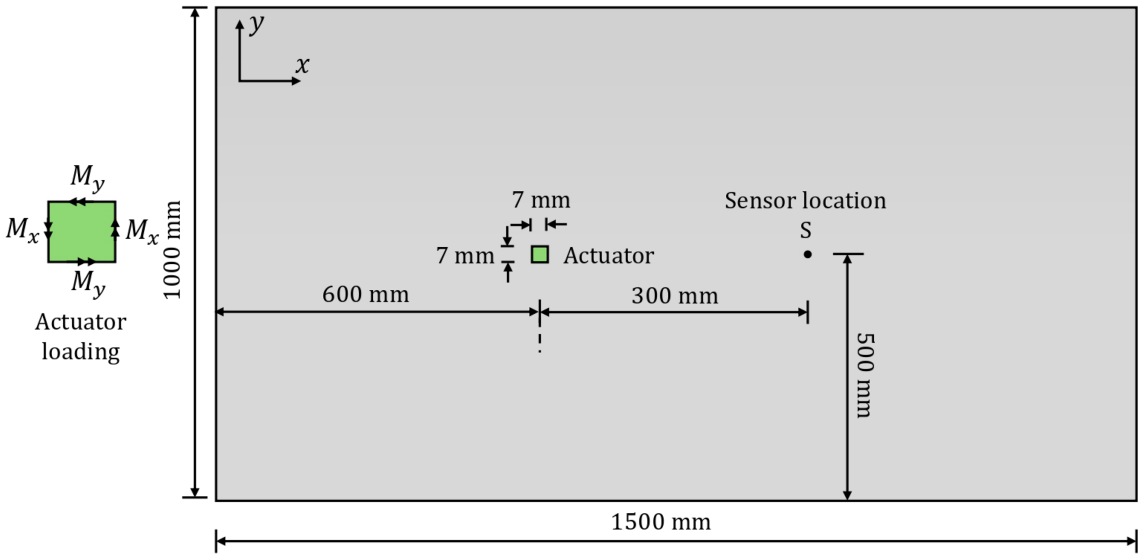
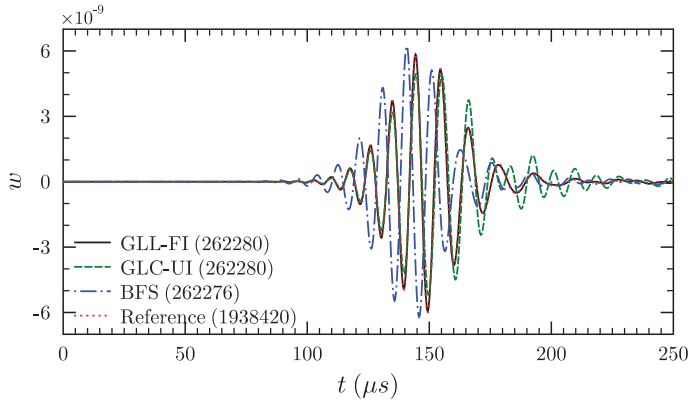
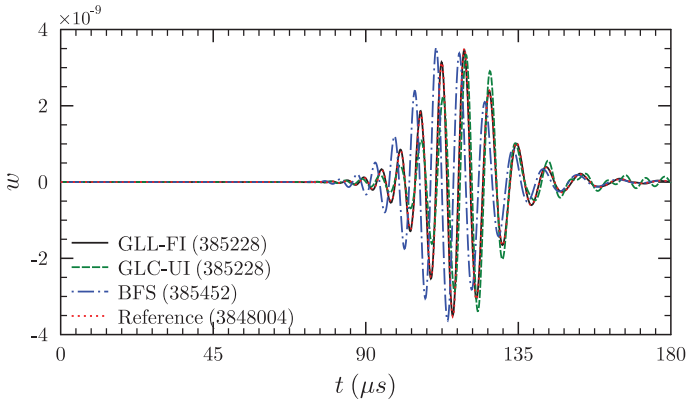


FIGURE 15 | Configuration of plate for patch case.



(a)



(b)

FIGURE 16 | Transverse displacement-time history at location S of the plate under 5-cycle tone burst excitation for central frequencies 100 and 150 kHz: Comparison of spectral and standard BFS elements. (a) 100 kHz; (b) 150 kHz.

step t_i obtained numerically and reference solutions, respectively. Furthermore, the convergence of the GLL-FI, GLL-UI* (DEM), and the standard BFS element solutions with increasing numbers of DOFs is illustrated in Figure 18. All the above different representations of the results indicate that the BFS element exhibits a significantly lower convergence rate compared to the current spectral formulation for both GLL-FI and GLL-UI* (DEM). On the other hand, GLL-FI yields the fastest convergence

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

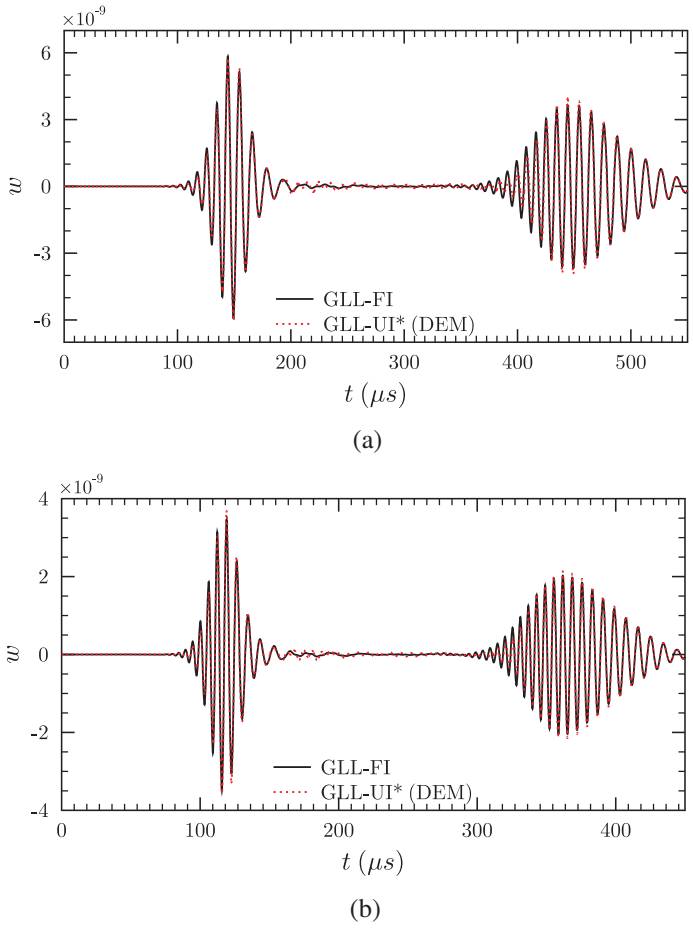


FIGURE 17 | Transverse displacement-time history showing edge reflections at location S on the top surface of the plate under 5-cycle tone burst excitation for excitation frequencies obtained with full- and under-integrated mass matrices. (a) 100 kHz; (b) 150 kHz.

TABLE 4 | Comparison of the accuracy of the present spectral plate element with the BFS element for wave propagation analysis.

Frequency (kHz)		Reference solution	GLL-FI	GLC-UI	BFS	GLL-UI* (DEM)
100	No. of DOFs	1,938,420	262,280	262,280	262,280	320,814
	RMSD (%)	0	1.84	32.3	179	6.42
150	No. of DOFs	3,848,004	385,228	385,228	385,228	479,098
	RMSD (%)	0	3.60	38.9	131	8.90

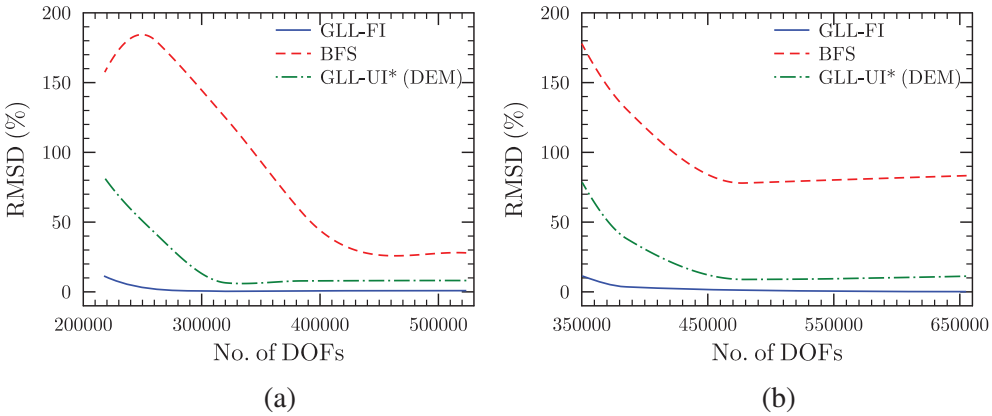


FIGURE 18 | Convergence of RMSD values of the response for excitation frequencies 100 and 150 kHz. (a) 100 kHz; (b) 150 kHz.

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and highest accuracy for the wave propagation problem. The GLL-UI* employing DEM, which is computationally the most efficient in terms of solution time (t_c), also yields comparable results with a percentage RMSD less than 9%. GLL-UI* (DEM) requires a lesser computational time of 0.7 times even with a larger number of DOFs compared to GLL-FI using implicit the Newmark-β time integration. The BFS element with an equivalent number of DOFs requires a computational time of 3 times that of GLL-FI while yielding much deviated solutions having a very high RMSD error. The GLC-UI of Wang et al. [99] too yields a much inferior solution of the wave propagation response with a significantly greater RMSD value compared to the present GLL element with full or under-integration of the mass matrix.

6 | Conclusions

The time-domain spectral BFS plate element is introduced for analyzing flexural guided wave propagation in thin isotropic plates. It utilizes a C^1-continuous spectral interpolation based on the modified Hermite polynomials, employing GLL points as the basis and selective collocation of DOFs. The lowest order version of the element reduces to the classical BFS plate element. The element is capable of diagonalizing the mass matrix using nodal quadrature, which enhances the computational efficiency by eliminating the need of matrix inversion through explicit time integration.Two methodologies are proposed to address issues arising from zero diagonals in the diagonalized mass matrix: Small value substitution (DEM) and Guyan reduction. Numerical investigations yield the following insights:

1. The proposed GLL basis spectral plate element provides accurate predictions for static and free vibration responses of plates with much higher accuracy than the existing elements, exhibiting optimal convergence rates.
2. For the present C^1-continuous interpolation too, the GLL points yield much lower values of condition numbers of system matrices than the EQ points, similar to the C^0-continuous shape functions. The condition numbers of the system matrices can be well fitted to the theoretical estimates available for C^0-continuous 2D elements with modified α-values.
3. The frequency spectra using the proposed element demonstrate improved accuracy for both lower and higher modes compared to existing plate elements, with deviations of less than 1%.
4. For lumped mass matrix formulation, the approximate DEM and the exact GRM for handling zero-diagonal terms exhibit comparable accuracy, with the latter being marginally superior. The GRM, however, takes larger computational time due to initial matrix operations.
5. The proposed element showcases exceptional accuracy, faster convergence, and superior computational efficiency compared to the existing GLC-based spectral element and the standard BFS element for high-frequency wave propagation problems in rectangular thin plates.

Despite its promising attributes, the element is limited to using only structured meshes, confining its use to plates with only rectangular subdomains. In future work, the proposed spectral Kirchhoff plate formulation will be extended to model thin plates with non-rectangular geometries using fictitious domain approaches [78, 79]. Further, the proposed C^1-continuous shape functions will be useful in developing other plate elements based on higher order theories requiring C^1-continuity for isotropic and composite plates.

Acknowledgments

S. Kapuria thankfully acknowledges the financial support provided by the Science & Engineering Research Board, Department of Science and Technology, Government of India, through the J. C. Bose National Fellowship (Grant No. JBR/2023/000025) and the Core Research Grant (CRG/2021/001237).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. J. L. Rose, *Ultrasonic Guided Waves in Solid Media* (New York: Cambridge University Press, 2014).

2. S. Kapuria, "Best Reconstruction Frequency for Time-Reversal Process of Lamb Waves and Its Determination From a Single Measurement," *Mechanical Systems and Signal Processing* 223 (2025): 111887.

3. J. J. Wang, T. P. Chang, B. T. Chen, and H. Wang, "Determination of Poisson's Ratio of Solid Circular Rods by Impact-Echo Method," *Journal of Sound and Vibration* 331, no. 5 (2012): 1059–1067.

4. A. Raghavan and C. Cesnik, "Review of Guided-Wave Structural Health Monitoring," *Shock and Vibration Digest* 39 (2007): 91–114, https://doi.org/10.1177/0583102406075428.

5. J. K. Agrahari and S. Kapuria, "A Refined Lamb Wave Time-Reversal Method With Enhanced Sensitivity for Damage Detection in Isotropic Plates," *Journal of Intelligent Material Systems and Structures* 27, no. 10 (2016): 1283–1305.

6. J. K. Agrahari and S. Kapuria, "Active Detection of Block Mass and Notch-Type Damages in Metallic Plates Using a Refined Time-Reversed Lamb Wave Technique," *Structural Control and Health Monitoring* 25, no. 2 (2018): e2064.

7. S. Kapuria and J. K. Agrahari, "Shear-Lag Solution for Excitation, Sensing, and Time Reversal of Lamb waves for structural health monitoring," *Journal of Intelligent Material Systems and Structures* 29, no. 4 (2018): 585–599.

8. M. Kannusamy, S. Kapuria, and S. Sasmal, "Accurate Baseline-Free Damage Localization in Plates Using Refined Lamb Wave Time-Reversal Method," *Smart Materials and Structures* 29, no. 5 (2020): 055044.

9. V. K. Chillara, J. Greenhall, and C. Pantea, "Ultrasonic Waves From Radial Mode Excitation of a Piezoelectric Disc on the Surface of an Elastic Solid," *Smart Materials and Structures* 29, no. 8 (2020): 085002.

10. W. Stronge, *Wave Propagation From Impact on Slender Deformable Bodies. ch. 7*. (Cambridge, UK: Cambridge University Press, 2018).

11. A. Riaud, J. L. Thomas, E. Charron, A. Bussonnière, O. B. Matar, and M. Baudoin, "Anisotropic Swirling Surface Acoustic Waves From Inverse Filtering for On-Chip Generation of Acoustic Vortices," *Physical Review Applied* 4, no. 3 (2015): 034004.

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

12. Z. Tian, S. Yang, P. H. Huang, et al., "Wave Number–Spiral Acoustic Tweezers for Dynamic and Reconfigurable Manipulation of Particles and Cells," *Science Advances* 5, no. 5 (2019): eaau6062.

13. J. Wang, S. Wang, L. Xie, et al., "The Axisymmetric Rayleigh Waves in a Semi-Infinite Elastic Solid," *Theoretical and Applied Mechanics Letters* 10, no. 2 (2020): 120–124.

14. J. Semblat and J. Brioist, "Efficiency of Higher Order Finite Elements for the Analysis of Seismic Wave Propagation," *Journal of Sound and Vibration* 231, no. 2 (2000): 460–467.

15. T. Sarai, T. Inoue, K. Fujiwara, and K. Kuramoto, "Dynamic Finite Element Analysis of Impulsive Stress Waves Propagating From Distal End of Femur," *Acta Medica Okayama* 66, no. 5 (2012): 409–415.

16. A. Ambardar and C. D. Ferris, "Wave Propagation in a Piezoelectric Two-Layered Cylindrical Shell With Hexagonal Symmetry: Some Implications for Long Bone," *Journal of the Acoustical Society of America* 63, no. 3 (1978): 781–792.

17. S. Gopalakrishnan, *Wave Propagation in Materials and Structures* (Boca Raton, FL: CRC Press, 2016).

18. C. Willberg, S. Duczek, J. V. Perez, D. Schmicker, and U. Gabbert, "Comparison of Different Higher Order Finite Element Schemes for the Simulation of Lamb Waves," *Computer Methods in Applied Mechanics and Engineering* 241 (2012): 246–261.

19. C. Willberg, S. Duczek, J. M. Vivar-Perez, and Z. A. Ahmad, "Simulation Methods for Guided Wave-Based Structural Health Monitoring: A Review," *Applied Mechanics Reviews* 67, no. 1 (2015): 010803.

20. Z. Su and L. Ye, *Identification of Damage Using Lamb Waves: From Fundamentals to Applications*, vol. 48 (London, UK: Springer Science & Business Media, 2009).

21. R. Lee and A. C. Cangellaris, "A Study of Discretization Error in the Finite Element Approximation of Wave Solutions," *IEEE Transactions on Antennas and Propagation* 40, no. 5 (1992): 542–549.

22. F. Moser, L. J. Jacobs, and J. Qu, "Modeling Elastic Wave Propagation in Waveguides With the Finite Element Method," *NDT & E International* 32, no. 4 (1999): 225–234.

23. S. Han, A. N. Palazotto, and C. L. Leakeas, "Finite-Element Analysis of Lamb Wave Propagation in a Thin Aluminum Plate," *Journal of Aerospace Engineering* 22, no. 2 (2009): 185–197.

24. J. Agrahari and S. Kapuria, "Effects of Adhesive, Host Plate, Transducer and Excitation Parameters on Time Reversibility of ULTRASONIC LAMB WAVES," *Ultrasonics* 70 (2016): 147–157.

25. M. Palacz, "Spectral Methods for Modelling of Wave Propagation in Structures in Terms of Damage Detection—A Review," *Applied Sciences* 8, no. 7 (2018): 1124.

26. L. Maio and P. Fromme, "On Ultrasound Propagation in Composite Laminates: Advances in Numerical Simulation," *Progress in Aerospace Sciences* 129 (2022): 100791.

27. S. K. Datta, A. H. Shah, R. Bratton, and T. Chakraborty, "Wave Propagation in Laminated Composite Plates," *Journal of the Acoustical Society of America* 83, no. 6 (1988): 2020–2026.

28. Z. Ahmad, J. M. Vivar-Perez, and U. Gabbert, "Semi-Analytical Finite Element Method for Modeling of LAMB WAVE PROPAGATION," *CEAS Aeronautical Journal* 4 (2013): 21–33.

29. M. K. Kalkowski, E. Rustighi, and T. P. Waters, "Modelling Piezoelectric Excitation in Waveguides Using the Semi-Analytical Finite Element Method," *Computers & Structures* 173 (2016): 174–186.

30. W. Duan and T. H. Gan, "Investigation of Guided Wave Properties of Anisotropic Composite Laminates Using a Semi-Analytical Finite Element Method," *Composites Part B: Engineering* 173 (2019): 106898.

31. E. Mirzaee Kakhki, J. Rezaeepazhand, F. Duvigneau, et al., "On the Use of High-Order Shape Functions in the SAFE Method and Their Performance in Wave Propagation Problems," *Mathematical and Computational Applications* 27, no. 4 (2022): 63.

32. B. R. Mace and E. Manconi, "Modelling Wave Propagation in Two-Dimensional Structures Using Finite Element Analysis," *Journal of Sound and Vibration* 318, no. 4–5 (2008): 884–902.

33. W. Zhou and M. Ichchou, "Wave Propagation in Mechanical Waveguide With Curved Members Using Wave Finite Element Solution," *Computer Methods in Applied Mechanics and Engineering* 199, no. 33–36 (2010): 2099–2109.

34. D. Chronopoulos, "Calculation of Guided Wave Interaction With Nonlinearities And Generation of Harmonics in Composite Structures Through a Wave Finite Element Method," *Composite Structures* 186 (2018): 375–384.

35. A. T. Patera, "A Spectral Element Method for Fluid Dynamics: Laminar Flow in a Channel Expansion," *Journal of Computational Physics* 54, no. 3 (1984): 468–488.

36. P. Kudela, A. Zak, M. Krawczuk, and W. Ostachowicz, "Modelling of Wave Propagation in Composite Plates Using the Time Domain Spectral Element Method," *Journal of Sound and Vibration* 302, no. 4–5 (2007): 728–745.

37. L. Ge, X. Wang, and F. Wang, "Accurate Modeling of PZT-Induced Lamb Wave Propagation in Structures by Using a Novel Spectral Finite Element Method," *Smart Materials and Structures* 23 (2014): 095018.

38. S. Eisenträger, S. Kapuria, M. Jain, and J. Zhang, "On the Numerical Properties of High-Order Spectral (Euler–Bernoulli) Beam Elements," *ZAMM—Journal of Applied Mathematics and Mechanics* 103 (2023): 1–45.

39. P. Kudela and W. M. Ostachowicz, "Wave Propagation Modelling in Composite Plates," *Applied Mechanics and Materials* 9 (2008): 89–104.

40. H. Peng, G. Meng, and F. Li, "Modeling of Wave Propagation in Plate Structures Using Three-Dimensional Spectral Element Method for Damage Detection," *Journal of Sound and Vibration* 320 (2009): 942–954.

41. Y. Kim, S. Ha, and F. K. Chang, "Time-Domain Spectral Element Method for Built-In Piezoelectric-Actuator-Induced Lamb Wave Propagation Analysis," *AIAA Journal* 46, no. 3 (2008): 591–600.

42. S. Ha and F. K. Chang, "Optimizing a Spectral Element for Modeling Pzt-Induced Lamb Wave Propagation in Thin Plates," *Smart Materials and Structures* 19, no. 1 (2010): 015015, https://doi.org/10.1088/0964-1726/19/1/015015.

43. A. Düster, H. Bröker, and E. Rank, "The p-version of the Finite Element Method for Three-Dimensional Curved Thin Walled Structures," *International Journal for Numerical Methods in Engineering* 52, no. 7 (2001): 673–703.

44. F. Li, H. Peng, X. Sun, J. Wang, and G. Meng, "Wave Propagation Analysis in Composite Laminates Containing a Delamination Using a Three-Dimensional Spectral Element Method," *Mathematical Problems in Engineering* 2012 (2012): 1–19.

45. K. Lonkar and F. K. Chang, "Modeling of Piezo-Induced Ultrasonic Wave Propagation in Composite Structures Using Layered Solid Spectral Element," *Structural Health Monitoring* 13, no. 1 (2014): 50–67.

46. J. F. Doyle, "A Spectrally Formulated Finite Element for Longitudinal Wave Propagation," *International Journal of Analytical and Experimental Modal Analysis* 3 (1988): 1–5.

47. J. F. Doyle, *Wave Propagation in Structures* (New York: Springer-Verlag, 1997).

48. D. R. Mahapatra and S. Gopalakrishnan, "A Spectral Finite Element Model for Analysis of Axial-Flexural-Shear Coupled Wave Propagation in Laminated Composite Beams," *Composite Structures* 59 (2003): 67–88.

49. S. Gopalakrishnan, M. Martin, and J. Doyle, "A Matrix Methodology for Spectral Analysis of Wave Propagation in Multiple Connected

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Timoshenko Beams," *Journal of Sound and Vibration* 158, no. 1 (1992): 11–24, https://doi.org/10.1016/0022-460X(92)90660-P.

50. N. Nanda, S. Kapuria, and S. Gopalakrishnan, "Spectral Finite Element Based on an Efficient Layerwise Theory for Wave Propagation Analysis of Composite and Sandwich Beams," *Journal of Sound and Vibration* 333 (2014): 3120–3137.

51. N. Nanda and S. Kapuria, "Spectral Finite Element for Wave Propagation in Curved Beams," *Journal of Vibration and Acoustics* 137 (2015): 41005.

52. N. Nanda and S. Kapuria, "Spectral Finite Element for Wave Propagation Analysis of Laminated Composite Curved Beams Using Classical and First Order Shear Deformation Theories," *Composite Structures* 132 (2015): 310–320.

53. S. Kapuria, "An Efficient Coupled Theory for Multilayered Beams With Embedded Piezoelectric Sensory and Active Layers," *International Journal of Solids and Structures* 38, no. 50–51 (2001): 9179–9199.

54. S. A. Rizzi and J. F. Doyle, "Spectral Analysis of Wave Motion in Plane Solids With Boundaries," *Journal of Vibration and Acoustics* 114, no. 2 (1992): 133, https://doi.org/10.1115/1.2930241.

55. S. A. Rizzi and J. F. Doyle, "A Spectral Element Approach to Wave Motion in Layered Solids," *Journal of Vibration and Acoustics* 114, no. 4 (1992): 569, https://doi.org/10.1115/1.2930300.

56. A. Chakraborty and S. Gopalakrishnan, "A Spectrally Formulated Finite Element for Wave Propagation Analysis in Layered Composite Media," *International Journal of Solids and Structures* 41, no. 18–19 (2004): 5155–5183, https://doi.org/10.1016/j.ijsolstr.2004.03.011.

57. A. Chakraborty and S. Gopalakrishnan, "A Spectrally Formulated Plate Element for Wave Propagation Analysis in Anisotropic Material," *Computer Methods in Applied Mechanics and Engineering* 194, no. 42–44 (2005): 4425–4446.

58. A. Chakraborty and S. Gopalakrishnan, "A Spectral Finite Element Model for Wave Propagation Analysis in Laminated Composite Plate," *Journal of Vibration and Acoustics* 128, no. 4 (2006): 477–488.

59. J. M. Melenk and I. Babuška, "The Partition of Unity Finite Element Method: Basic Theory and Applications," *Computer Methods in Applied Mechanics and Engineering* 139, no. 1–4 (1996): 289–314.

60. O. Laghrouche, P. Bettess, and R. Astley, "Modelling of Short Wave Diffraction Problems Using Approximating Systems of Plane Waves," *International Journal for Numerical Methods in Engineering* 54, no. 10 (2002): 1501–1533.

61. A. El Kacimi and O. Laghrouche, "Improvement of PUFEM for the Numerical Solution of High-Frequency Elastic Wave Scattering on Unstructured Triangular Mesh Grids," *International Journal for Numerical Methods in Engineering* 84, no. 3 (2010): 330–350.

62. A. Kumar and S. Kapuria, "An Enriched Finite Element Method for General Wave Propagation Problems Using Local Element Domain Harmonic Enrichment Functions," *Archive of Applied Mechanics* 88 (2018): 1573–1594.

63. S. Kapuria and A. Kumar, "A Wave Packet Enriched Finite Element for Electroelastic Wave Propagation Problems," *International Journal of Mechanical Sciences* 170 (2020): 105081.

64. M. Hrabok and T. Hrudey, "A Review and Catalogue of Plate Bending Finite Elements," *Computers & Structures* 19, no. 3 (1984): 479–495.

65. A. Adini and R. W. Clough, *Analysis of Plate Bending by the Finite Element Method* (Berkeley, CA: University of California, 1960).

66. R. J. Melosh, "Basis for Derivation of Matrices for the Direct Stiffness Method," *AIAA Journal* 1, no. 7 (1963): 1631–1637.

67. O. C. Zienkiewicz and Y. K. Cheung, "The Finite Element Method for Analysis of Elastic Isotropic and Orthotropic Slabs," *Proceedings of the Institution of Civil Engineers* 28, no. 4 (1964): 471–488.

68. R. W. Clough, "Finite Element Stiffness Matricess for Analysis of Plate Bending," in *Proceedings of the First Conference on Matrix Methods in Struct. Mech* (1965), 515–546.

69. J. H. Argyris, I. Fried, and D. W. Scharpf, "The TUBA Family of Plate Elements for the Matrix Displacement Method," *Aeronautical Journal* 72, no. 692 (1968): 701–709.

70. K. Bell, "A Refined Triangular Plate Bending Finite Element," *International Journal for Numerical Methods in Engineering* 1, no. 1 (1969): 101–122.

71. B. M. Irons, "A Conforming Quartic Triangular Element for Plate Bending," *International Journal for Numerical Methods in Engineering* 1, no. 1 (1969): 29–45.

72. R. W. Clough and C. A. Felippa, "A Refined Quadrilateral Element for Analysis of Plate Bending," in *Proceedings of the Wright-Patterson Air Force Base* (1968), 399.

73. B. F. De Veubeke, "A Conforming Finite Element for Plate Bending," *International Journal of Solids and Structures* 4, no. 1 (1968): 95–108.

74. F. K. Bogner, R. L. Fox, and L. A. Schmit, "The Generation of Interelement Compatible Stiffness and Mass Matrices by the Use of Interpolation Formulae," in *Proceedings of the Conference on Matrix Methods in Structural Mechanics, Wright Patterson AF Base, Ohio* (1965).

75. O. C. Zienkiewicz, R. L. Taylor, and J. Z. Zhu, *The Finite Element Method: Its Basis and Fundamentals* (Oxford, UK: Elsevier, 2005).

76. S. Gopalacharyulu, "A Higher Order Conforming, Rectangular Plate Element," *International Journal for Numerical Methods in Engineering* 6, no. 2 (1973): 305–308.

77. D. S. Watkins, "On the Construction of Conforming Rectangular Plate Elements," *International Journal for Numerical Methods in Engineering* 10, no. 4 (1976): 925–933.

78. S. Eisenträger, J. Kiendl, G. Michaloudis, R. Duy, and Y. Vetyukov, "Stability Analysis of Plates Using Cut Bogner-Fox-Schmit Elements," *Computers & Structures* 270 (2022): 106854.

79. E. Burman, P. Hansbo, and M. G. Larson, "Cut Bogner-Fox-Schmit Elements for Plates," *Advanced Modeling and Simulation in Engineering Sciences* 7, no. 1 (2020): 27.

80. J. Kiendl, K. U. Bletzinger, J. Linhard, and R. Wüchner, "Isogeometric Shell Analysis With Kirchhoff–Love Elements," *Computer Methods in Applied Mechanics and Engineering* 198, no. 49–52 (2009): 3902–3914.

81. T. J. Hughes, J. A. Cottrell, and Y. Bazilevs, "Isogeometric Analysis: Cad, Finite Elements, Nurbs, Exact Geometry and Mesh Refinement," *Computer Methods in Applied Mechanics and Engineering* 194, no. 39–41 (2005): 4135–4195.

82. J. A. Cottrell, A. Reali, Y. Bazilevs, and T. J. Hughes, "Isogeometric Analysis of Structural Vibrations," *Computer Methods in Applied Mechanics and Engineering* 195, no. 41–43 (2006): 5257–5296.

83. O. Zienkiewicz, R. Taylor, and J. Too, "Reduced Integration Technique in General Analysis of Plates and Shells," *International Journal for Numerical Methods in Engineering* 3, no. 2 (1971): 275–290.

84. T. J. Hughes, M. Cohen, and M. Haroun, "Reduced and Selective Integration Techniques in the Finite Element Analysis of Plates," *Nuclear Engineering and Design* 46, no. 1 (1978): 203–222.

85. K. J. Bathe and E. N. Dvorkin, "A Four-Node Plate Bending Element Based on Mindlin/Reissner Plate Theory and a Mixed Interpolation," *International Journal for Numerical Methods in Engineering* 21, no. 2 (1985): 367–383.

86. J. L. Batoz and M. B. Tahar, "Evaluation of a New Quadrilateral Thin Plate Bending Element," *International Journal for Numerical Methods in Engineering* 18, no. 11 (1982): 1655–1677.

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

87. K. U. Bletzinger, M. Bischoff, and E. Ramm, "A Unified Approach for Shear-Locking-Free Triangular and Rectangular Shell Finite Elements," *Computers & Structures* 75, no. 3 (2000): 321–334.

88. S. Cen and Y. Shang, "Developments of Mindlin-Reissner Plate Elements," *Mathematical Problems in Engineering* 2015 (2015): 456740.

89. U. Zrahia and P. Bar-Yoseph, "Plate Spectral Elements Based Upon Reissner–Mindlin Theory," *International Journal for Numerical Methods in Engineering* 38, no. 8 (1995): 1341–1360.

90. K. D. Brito and M. A. Sprague, "Reissner–Mindlin Legendre Spectral Finite Elements With Mixed Reduced Quadrature," *Finite Elements in Analysis and Design* 58 (2012): 74–83.

91. H. Dang-Trung, D. J. Yang, and Y. Liu, "Improvements in Shear Locking and Spurious Zero Energy Modes Using Chebyshev Finite Element Method," *Journal of Computing and Information Science in Engineering* 19, no. 1 (2019): 011006.

92. C. P. Fritzen, R. Schulte, and H. Jung, "A Modelling Approach for Virtual Development of Wave Based SHM systems," *Journal of Physics: Conference Series* 305 (2011): 12–71.

93. R. T. Schulte and C. P. Fritzen, "Simulation of Wave Propagation in Damped Composite Structures With Piezoelectric Coupling," *Journal of Theoretical and Applied Mechanics* 49, no. 3 (2011): 879–903.

94. C. S. Rekatsinas and D. A. Saravanos, "A Time Domain Spectral Layerwise Finite Element for Wave Structural Health Monitoring in Composite Strips With Physically Modeled Active Piezoelectric Actuators and Sensors," *Journal of Intelligent Material Systems and Structures* 28, no. 4 (2017): 488–506.

95. C. S. Rekatsinas and D. A. Saravanos, "A Cubic Spline Layerwise Time Domain Spectral Fe for Guided Wave Simulation in Laminated Composite Plate Structures With Physically Modeled Active Piezoelectric Sensors," *International Journal of Solids and Structures* 124 (2017): 176–191.

96. S. Kapuria and M. Jain, "A C^1-Continuous Time Domain Spectral Finite Element for Wave Propagation Analysis of Euler–Bernoulli Beams," *International Journal for Numerical Methods in Engineering* 122 (2021): 2631–2652.

97. L. Han, J. Wang, H. Li, and G. Sun, "A Time-Domain Spectral Element Method With C^1 Continuity for Static and Dynamic Analysis of Frame Structures," *Structure* 28 (2020): 604–613.

98. M. Jain, S. Kapuria, and S. Pradyumna, "Efficient Time-Domain Spectral Element With Zigzag Kinematics for Multilayered Strips," *International Journal of Mechanical Sciences* 232 (2022): 107–603.

99. J. Wang, H. Li, G. Sun, and L. Han, "Free Vibration Analysis of Rectangular Thin Plates With Corner and Inner Cutouts Using C^1 Chebyshev Spectral Element Method," *Thin-Walled Structures* 181 (2022): 110031.

100. C. Pozrikidis, *Introduction to Finite and Spectral Element Methods using MATLAB* (Boca Raton, FL: CRC press, 2005).

101. L. Radtke, D. Müller, and A. Düster, "Optimally Blended Spectral Elements in Structural Dynamics: Selective Integration and Mesh Distortion," *International Journal of Computational Methods* 18, no. 10 (2021): 2150042.

102. W. Ostachowicz, P. Kudela, M. Krawczuk, and A. Zak, *Guided Waves in Structures for SHM: The Time-Domain Spectral Element Method* (Chichester: John Wiley & Sons, 2011).

103. S. Timoshenko and S. Woinowsky-Krieger, *Theory of Plates and Shells. 2* (New York: McGraw-Hill New York, 1959).

104. S. S. Rao, *Vibration of Continuous Systems* (Hoboken, NJ: John Wiley & Sons, 2019).

105. J. F. Maitre and O. Pourquier, "Condition Number and Diagonal Preconditioning: Comparison of the p-version and the Spectral Element Methods," *Numerische Mathematik* 74, no. 1 (1996): 69–84.

106. J. M. Melenk, "On Condition Numbers in Hp-Fem With Gauss–Lobatto-Based Shape Functions," *Journal of Computational and Applied Mathematics* 139, no. 1 (2002): 21–48.

107. P. Gervasio, L. Dedè, O. Chanon, and A. Quarteroni, "A Computational Comparison Between Isogeometric Analysis and Spectral Element Methods: Accuracy and Spectral Properties," *Journal of Scientific Computing* 83 (2020): 1–45.

108. V. Giurgiutiu, *Structural Health Monitoring: With Piezoelectric Wafer Active Sensors* (Oxford: Elsevier, 2007).

109. G. Sharma and J. Martin, "MATLAB: A Language for Parallel Computing," *International Journal of Parallel Programming* 37 (2009): 3–36.

110. J. Kepner, *Parallel MATLAB for Multicore and Multinode Computers* (Philadelphia, PA: SIAM, 2009).

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix A

MATLAB Code for High-Order BFS Shape Functions With Selective Collocation of Dofs

MATLAB Function to Obtain Shape Functions

```
% SF_BFS.m
% Compute the function vales of the Bogner--Fox--Schmit shape functions
%% Input
% (xi,eta):     Local coordinates (have to be provided as vectors)
%         n:     Number of nodes per edge
%  la & lb:     Length and width of the plate
%        SF:     Shape functions obtained using GLC or GLL roots
%% Output
% N:            Matrix of shape functions evaluated at (xi,eta)
function [N] = SF_BFS(xi,eta,n,la,lb,SF)

% Number of points for basis
No = n-1;
        if SF ==1
% GLC roots using lgcnodes.m
                [x,~] = lgcnodes(No);
        elseif SF==2
% GLL roots using lglnodes.m
                [x,~] = lglnodes(No);
        end
% To obtain Lagrange and Hermite polynomials as per formulation using
% Lagrange_SF_xi.m, Lagrange_SF_eta.m
% Hermite_SF_1xi.m, Hermite_SF_2xi.m, Hermite_SF_1eta and Hermite_SF_2eta
        [L_xi,L_dxi,L_dxi2,L_dxii] = Lagrange_SF_xi(n,x,xi);
        [H0_xi] = Hermite_SF_1xi(L_xi,L_dxi,L_dxi2,L_dxii,x,n,xi);
        [H1_xi] = Hermite_SF_2xi(L_xi,L_dxi,L_dxi2,x,n,xi);
        [L_eta,L_deta,L_deta2,L_detai] = Lagrange_SF_eta(n,x,eta);
        [H0_eta] = Hermite_SF_1eta(L_eta,L_deta,L_deta2,L_detai,x,n,eta);
        [H1_eta] = Hermite_SF_2eta(L_eta,L_deta,L_deta2,x,n,eta);

% initialize shape functions
N=zeros(n^2,4);
% Tensor product of 1D Hermite polynomials (Equation (28)) corresponding
% to 4-DOFs per node
% Node numbering as per Figure (2) using Node_no_SF.m
        for i=1:n
                for j=1:n
                        Node_no=Node_no_SF(i,j,n);
        % For DOF w
                        N(Node_no, 1) = H0_xi(i) * H0_eta(j);
        % For DOF theta_x
                        if (i==1 || i==n)
                                N(Node_no, 2) = (la/2)*H1_xi(i) * H0_eta(j);
                        end
        % For DOF theta_y
                        if (j==1 || j==n)
                                N(Node_no, 3) = (lb/2)*H0_xi(i) * H1_eta(j);
                        end
        % For DOF theta_xy
                        if (i==1 || i==n) && (j==1 || j==n)
                                N(Node_no, 4) = (la*lb/4)*H1_xi(i) * H1_eta(j);
                        end
                end
        end
N = N.';
% To column vector
N = reshape(N,1,[]);
N = N.';
% Eliminating rows of DOFs which are not required that have no function value (Equation (51)
% and Figure (2))
l1 = 18:4:(16+4*(n-2)); % edge 1 (bottom) thetax
```


10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

```
l2 = 20:4:(16+4*(n-2)); % edge 1 wxy
l3 = (16+4*(n-2))+3:4:(16+8*(n-2));  % edge 2 thetay
l4 = (16+4*(n-2))+4:4:(16+8*(n-2));  % edge 2 wxy
l5 = (16+8*(n-2))+2:4:(16+12*(n-2)); % edge 3 thetax
l6 = (16+8*(n-2))+4:4:(16+12*(n-2)); % edge 3 wxy
l7 = (16+12*(n-2))+3:4:(16+16*(n-2)); % edge 4 thetay
l8 = (16+12*(n-2))+4:4:(16+16*(n-2)); % edge 4 wxy
l9 = (16+16*(n-2))+2:4:(4*(n)^2); % mid node thetax
l10 = (16+16*(n-2))+3:4:(4*(n)^2); % mid node thetay
l11 = (16+16*(n-2))+4:4:(4*(n)^2); % mid node wxy
l=[l1 l2 l3 l4 l5 l6 l7 l8 l9 l10 l11];
N(l,:)=[];
end
```

MATLAB Function to Obtain Hermite Polynomials (H_{0i}) and (H_{1i}) and Its Derivatives

```
% Hermite_SF_1xi.m
function [H0_xi,H0_dxi,H0_dxi2] = Hermite_SF_1xi(L_xi,L_dxi,L_dxi2,L_dxii,x,n,xi)
% Initialize
H0_xi = zeros(1,n);
H0_dxi = zeros(1,n);
H0_dxi2 = zeros(1,n);
a =zeros(1,n);
b=zeros(1,n);
c=zeros(1,n);

% Modified Hermite polynomials (Equation (46))
for i = 1:n
    if i==1
        a(i) = (-0.5*x(i)*L_dxii(i) - 0.25);
        b(i) = 0.5*x(i);
        c(i) = (0.75 + 0.5*x(i)*L_dxii(i));
        H0_xi(i)   = (a(i)*xi^2 + b(i)*xi + c(i))*L_xi(i);
        H0_dxi(i)  = (2*a(i)*xi + b(i))*L_xi(i) + (a(i)*xi^2 + b(i)*xi + c(i))*L_dxi(i);
        H0_dxi2(i) = 2*a(i)*L_xi(i) + 2*(2*a(i)*xi +b(i))*L_dxi(i) + (a(i)*xi^2 + b(i)*xi +
                     c(i))*L_dxi2(i);
    elseif i==n
        a(i) = (-0.5*x(i)*L_dxii(i) - 0.25);
        b(i) = 0.5*x(i);
        c(i) = (0.75 + 0.5*x(i)*L_dxii(i));
        H0_xi(i) = (a(i)*xi^2 + b(i)*xi + c(i))*L_xi(i);
        H0_dxi(i) = (2*a(i)*xi + b(i))*L_xi(i) + (a(i)*xi^2 + b(i)*xi + c(i))*L_dxi(i);
        H0_dxi2(i) = 2*a(i)*L_xi(i) + 2*(2*a(i)*xi +b(i))*L_dxi(i) + (a(i)*xi^2 + b(i)*xi
                     + c(i))*L_dxi2(i);
    else
        H0_xi(i) = ((xi^2 -1)/(x(i)^2 - 1))*L_xi(i);
        H0_dxi(i) = (2*xi/(x(i)^2 - 1))*L_xi(i) + ((xi^2-1)/(x(i)^2-1))*L_dxi(i);
        H0_dxi2(i) = (2/(x(i)^2 - 1))*(L_xi(i) + 2*xi*L_dxi(i)) +
                                        ((xi^2-1)/(x(i)^2-1))*L_dxi2(i);
    end
end
end

% Hermite_SF_2xi.m
function [H1_xi,H1_dxi,H1_dxi2] = Hermite_SF_2xi(L_xi,L_dxi,L_dxi2,x,n,xi)
H1_xi = zeros(1,n);
H1_dxi = zeros(1,n);
H1_dxi2 = zeros(1,n);
% Modified Hermite polynomials (Equation (46))
for i=1:n
        if i ==1 || i==n
                H1_xi(i) = 0.5*((xi^2 -1)*x(i))*L_xi(i);
                H1_dxi(i) = x(i)*xi*L_xi(i) +0.5*(xi^2-1)*x(i)*L_dxi(i);
                H1_dxi2(i) = x(i)*L_xi(i) + 2*x(i)*xi*L_dxi(i) + 0.5*(xi^2-1)*x(i)*L_dxi2(i);
        end
end
end
```


10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

```
% Lagrange_SF_xi.m
function [L_xi,L_dxi,L_dxi2,L_dxii] = Lagrange_SF_xi(n,x,xi)
L_xi = zeros(1,n);
L_dxi = zeros(1,n);
L_dxi2 = zeros(1,n);
for i=1:n
    L1=1;
    for j=1:n
        if(j==i)
            l=0;
        else
            L1=L1*(xi-x(j))/(x(i)-x(j));
        end
    end
    L_xi(i)=vpa(L1);
end
for i=1:n
    sum1=0;
    for m=1:n
         L1=1;
     for j=1:n
       if(j==i || j==m)
           l=0;
       else
       L1=L1*(xi-x(j))/(x(i)-x(j));
       end
     end
         if (m==i)
         l=0;
     else
         sum1 = sum1 + (L1/(x(i)-x(m)));
     end
    end
         L_dxi(i)=sum1;
end
L_dxiisubs = zeros(n,n);
L_dxii = zeros(1,n);
for k=1:n
for i=1:n
    sum1=0;
    for m=1:n
         L1=1;
     for j=1:n
       if(j==i || j==m)
          l=0;
       else
       L1=L1*(x(k)-x(j))/(x(i)-x(j));
       end
     end
         if (m==i)
         l=0;
     else
         sum1 = sum1 + (L1/(x(i)-x(m)));
     end
    end
    L_dxiisubs(i,k)=sum1;
end
L_dxii(1,k)= L_dxiisubs(k,k);
end
for j=1:n
    sum1=0;
    for l=1:n
        if (j~=l)
            sum2=0;
        for w=1:n
            if (w~=j && w~=l)
```


10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

```
            L1=1;
            for m=1:n
                if(m~=j && m~=l && m~=w)
                    L1=L1*(xi-x(m))/(x(j)-x(m));
                end
            end
            sum2 = sum2 + (L1/(x(j)-x(w)));
            end
        end
            sum1 = sum1 + sum2/(x(j)-x(l));
        end
    end
         L_dxi2(j)=sum1;
end
end
```

10970207, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/nme.7617 by Universitaet Magdeburg, Wiley Online Library on [02/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License